

Experimental apparatus for universal spin-motion control in multi-ion registers



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A thesis submitted for the degree of
Doctor of Philosophy in Physics

Trinity 2026

Contents

0.1	Overview	1
0.2	Motivation	5
1	Theory	9
1.1	Trapping Potential	9
1.2	Light Matter Interactions	10
1.3	Light Matter Interactions	10
2	Ion Trap Apparatus	11
2.1	Specification	12
2.2	The Ion Trap	14
2.2.1	Trap RF Chain	16
2.2.2	Trap DC Voltages	21
2.2.3	System Grounding	22
2.3	Magnetic Field	22
2.4	Vacuum System and Beam Geometry	23
2.4.1	Trap Package	26
2.4.2	Beam Geometry	26
2.5	Imaging System	26
2.6	Ca ⁺ Laser Systems	26
2.6.1	Diode Lasers	27
2.6.2	Narrow Linewidth 729 Laser	27
2.6.3	Global Addressing System	31
2.6.4	Single Addressing System	31
2.7	Control System	33
2.8	Ion Loading	33
3	Experimental Characterisation	34
3.1	The Ion	36
3.2	Spin	39
3.2.1	State Preparation	40
3.2.2	Measurement	41
3.2.3	Single Qubit Rotations	42

	Gate Time Calibration	43
	Randomised Benchmarking	44
3.2.4	Spin Coherence Times	45
3.3	Motion	48
3.3.1	Cooling	48
	Doppler Cooling	49
	Sideband Cooling	51
3.3.2	Heating Rates	52
3.3.3	Motional Mode Stability	53
3.3.4	Motional Coherence Times	55
3.4	Spin-Motion	57
3.4.1	Spin-Dependent Forces	57
	Calibrating the SDF	58
3.4.2	Two-Qubit Entangling Gates	61

Appendices

A	Appendix	66
A.1	Generating Ions	66
A.2	Extracting Laser Offset and Magnetic Field	66
A.3	Experimental Control	66
A.4	Creating Squeezed States	66

References	67
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0.1 Overview

Proposed thesis structure

1. Introduction/Theory

- Ion trap theory needed.
 - Trapping fields
 - QP transitions + group theory on this
 - Ion-light interactions. With spin, with motion.
 - MS gates + SDFs
- Noise (Look at Harty + Amy thesis)
 - Model for general noise. Using error channels + PTM + Choi matrices. Discuss classifications of noise channels.
 - How to measure noise. Tomography, RB.
 - How to simplify noise using RB.
 - How k-copy RB simplifies noise but retains structure.
 - Group + representation theory for RB
 - Reducing physical noise. Coherent with H engineering, incoherent with fast gates.
 - Reducing the impact of noise by error mitigation techniques.
 - Virtual distillation on spatially correlated noise.

2. Experimental Apparatus – Description of the new experimental apparatus.

- Intro for why we built a new apparatus. What we wanted to achieve with it.
- Block diagram of whole experiment highlighting automation.
- See current chapter for most sections.
- Ground loops and how we found them. Suggest improvements to current system.

- Single addressing system. Chat to IRO about what minimal info to include.
- Section on Pulse shaping with phaser, sinara? Link this to requirements/theory for frequency selectivity.
- Verdi noise spectra
- 729 laser characterisation. Including linewidth, cavity ringdown, noise spectra. FNC noise reduction. Keeping it alive? Autolock feature.

3. **Experimental Characterisation** – spin/motion/spin-motion elementary interactions that are used throughout the thesis. This is the chapter submitted for confirmation of status.

- Spin coherence times with and without box, and with and without echo sequence.
- 729 laser characterisation. Including linewidth, cavity ringdown, noise spectra. FNC noise reduction. Keeping it alive?
- New mode stability with enclosed resonator
- Motional coherence time
- Going to long chains
- micromotion compensation
- All mode frequencies. Using James.
- All modes SBC. Here also a discussion on sub-doppler cooling options.
- MS gates improved + Walshing
- Error budget for MS gates
- QP matrix elements -> polarisations -> Stark shifts?

4. **k-copy Randomised Benchmarking** – Presentation of the theory and experimental results demonstrating parallel single-qubit randomised benchmarking to distinguish local and global noise contributions on each qubit. This is the paper extended.

- How k-copy RB simplifies noise but retains structure.
- Group + representation theory for k-copy RB to derive correlated populations
- Shared global fields leading to spatially correlated noise. Error channel for this.
- Optimal sampling for multi-exponential
- Parameter range
- Noise injection calibration
- Stark shift calibration
- Extracting errors for large parameter space
- Parity measurements
- Pairwise measurements

5. **Motional Control** – Presentation of the experimental results demonstrating motional control

- Describe Two-SDF toolkit
- Describe tomography method for motional states. Could show Fock state here.
- Results:
 - Squeezed state
 - Beam splitter
 - Parity operation
 - Fredkin gate

6. **Virtual Distillation** – Presentation of the experimental results demonstrating virtual distillation.

- two copy, B matrix demonstration
- Virtual distillation on spatially correlated noise
- any-to-any two-qubit gate
- $\pi/4$ calibration
- Motional virtual distillation. What errors does this mitigate? How to implement it? Results?

7. **Conclusion and Outlook**

0.2 Motivation

The following chapter details the characterisation work performed during the commissioning of a new ion trap apparatus. Over the past two years, my colleagues and I have been constructing this system, which is now approaching readiness for our desired experiments. In this *Motivation* section, I will provide a high-level overview of the physics we want to explore, state the unique aspects of the apparatus that enable these experiments, and finally, give a timeline for planned future work. The complexity of our system necessitates collaboration, and so I would like to acknowledge the rest of the *FastGates* team, for their extensive contributions to bringing the system online.

There are two research directions that we are pursuing with this apparatus. Both involve the interplay between the “spin” electronic degrees of freedom and the motion of the ions which we describe by quantum harmonic oscillators.

The first is the demonstration of fast, high-fidelity two-qubit entangling gates between trapped ions. High-fidelity entangling gates are critical for developing a quantum computer which can perform useful calculations with high success probability [1]. Gate errors associated with decoherence of the quantum state are often dominated by incoherent processes, such as spontaneous emission, or interactions between the qubit and the environment [2]. These errors accumulate with time, and so may be mitigated by increasing the algorithm “clock-speed”, hence our desire for fast entanglement [3, 4]. Typically in ion traps, local two-qubit gates are performed via the ion’s shared motion, which may be excited by interaction with a laser [5, 6]. As the effective Rabi frequency approaches that of the frequency separation of the motional modes, the dynamics between spin and motion become complicated by multiple simultaneous mode excitations [4]. Our apparatus is designed to provide the required laser intensities and mode frequencies such that we can explore this regime, and demonstrate the required control to tame these multiple mode excitations and perform high-fidelity entangling gates.

The second direction is the use of the motion of the ion directly as a quantum

resource [7, 8]. This falls into the continuous variable quantum computing paradigm (CVQC), where instead of discrete qubits, the quantum state is described by bosonic “qumodes” [9]. CVQC reduces the technical overhead of scaling discrete systems, as larger Hilbert spaces are easily accessed due to the many level nature of the individual qumodes. We want to demonstrate a universal set of gates for CVQC, which includes, single-mode Gaussian and non-Gaussian gates (demonstrated in [10]), and two-mode entangling gates. Further, we want to demonstrate the ability to simultaneously leverage both bosonic and spin degrees of freedom of our ions to efficiently encode problems in the full hybrid system we have access to [11, 12]. Our apparatus has both precise motional control and stability through our trap choice, and the individual addressing system that is required when interacting with many-ion chains.

Details of the apparatus are presented in the Appendices. Summarised here are the unique aspects enabling the desired experiments:

- **High laser intensities:** The system is designed to provide high laser intensities, which are required to drive fast entangling gates. This is achieved through the use of a 5 W Ti:Sapph system (section ??) and two high numerical aperture ($\text{NA} = 0.6$) objectives (section 2.4).
- **Single ion addressing:** Using crossed acousto-optic deflectors, we can selectively address individual ions in a multi-ion chain (section 2.6.4). This enables fast entangling gates between ions in a long chain, and the ability to perform motional interactions on selected modes.
- **Standing wave addressing:** The system is designed to provide standing optical waves at the ion position. Standing waves are required for some fast gate schemes [13].
- **Motional control:** The Paul trap used in the system, produced by *National Physical Laboratory* in the UK, has low heating rates (section 3.3.2), deep trapping potentials relative to surface style ion traps, and flexible control

over DC confinement (section 2.2) (e.g. ion shuttling). Low heating rates are required for long motional coherence times (section 3.3.3).

Here, the characterisation of the spin qubits is in the context of discrete variable quantum computing. We are interested in the fidelity of qubit state-preparation (section 3.2.1), measurement (section 3.2.2), single- (section 3.2.3) and two-qubit gates (section 3.4.2). These individual elements comprise longer circuits, and so all contribute to the overall fidelity of any chosen algorithm.

The motional characterisation is currently focused on the stability of the modes (section 3.3.3), and the ability to perform spin-dependent forces, SDFs (section 3.4.1). SDFs are the base interaction on which many of the desired CVQC gates are comprised [14], and so these must be well characterised and understood in our system.

To push the system forwards, in the next 6 months, we will focus on commissioning the single addressing system, and on demonstrating squeezing of the motional state of the ion.

As described above, the single addressing system is a critical component for many of our planned experiments. Section 2.6.4, and [15] describe the system in detail. The immediate next steps are to align the constructed optical system to the ion chain, and to implement the logical and RF control system required to drive the acousto-optic deflectors. Once this is complete, we will characterise the crosstalk of the system by performing single addressing of ions in a chain and measuring spin populations. We will then be able to demonstrate fast entangling gates between ions in a chain, and the ability to perform motional interactions on selected modes. Squeezing of the motional mode via the method outlined in [14], is a significant step towards demonstrating the universal set of gates for CVQC. The constituent interactions of the effective squeezing gate (two SDFs), are the same interactions needed for trisqueezing [10, 16] — a non-Gaussian gate, and for two-mode squeezing [17] — an entangling gate. Furthermore, once we have demonstrated generating squeezing characteristic functions, there is a clear path to experimentally demonstrating the

extraction of Feynman diagram terms with the hybrid system being used as a quantum simulator [11].

1

Theory

Contents

1.1	Trapping Potential	9
1.2	Light Matter Interactions	10
1.3	Light Matter Interactions	10

1.1 Trapping Potential

From Earnshaw's theorem, $\nabla^2 V = 0$, a stable stationary point in 3D cannot be realised using only static electric potentials, V , as if the potential is confining in two dimensions, it will be anti-confining in the third. Therefore, to achieve stable trapping, either an oscillating electric field (Paul trap [18]), or a static magnetic field (Penning trap) is used.

Will include pseudopotential relating V_{RF} and Ω_{RF} to the radial mode frequencies ω_m (see Vera eqn 3.4),

$$\omega_r \propto \frac{V_{RF}}{M\Omega_{RF}}. \quad (1.1)$$

Further, will use James to give ion spacing and non-COM mode frequencies as will be relevant for single addressing, fast gates (if discussed), and for motional

mode excitations will be relevant for single addressing, fast gates (if discussed), and for motional mode excitation.

1.2 Light Matter Interactions

1.3 Light Matter Interactions

2

Ion Trap Apparatus

Contents

2.1	Specification	12
2.2	The Ion Trap	14
2.2.1	Trap RF Chain	16
2.2.2	Trap DC Voltages	21
2.2.3	System Grounding	22
2.3	Magnetic Field	22
2.4	Vacuum System and Beam Geometry	23
2.4.1	Trap Package	26
2.4.2	Beam Geometry	26
2.5	Imaging System	26
2.6	Ca⁺ Laser Systems	26
2.6.1	Diode Lasers	27
2.6.2	Narrow Linewidth 729 Laser	27
2.6.3	Global Addressing System	31
2.6.4	Single Addressing System	31
2.7	Control System	33
2.8	Ion Loading	33

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After 70 years of ion confinement development (since the quadrupole mass spectrometer in the 1950s [18]), and the myriad use-cases from time standards, to analytic chemistry, to quantum computing, there are naturally many systems and thesis apparatus chapters in existence today. As such, we structure this chapter in the

following way: We start with the standard requirements of an ion trap system and introduce the unique motivation and design specifications of our system; We then present an abstract schematic of the system in its current form, highlighting the constituent subsystems, and the optical and electrical signal paths; The remaining sections of the chapter contain the technical detail, perhaps more relevant to the practicing experimental student, and the hardware enthusiastic.

2.1 Specification

Ion-trap experiments are exceptionally versatile in their application, however in the vast search space of quantum-information applications, we want a system particularly tailored for three avenues of study:

1. Error channels on multi-qubit chains.
2. Continuous variable quantum computing and simulation utilising the spin and motional degrees of freedom of the confined ions.
3. Fast two-qubit entangling gates via phase stabilised optical standing waves.

These latter two research directions were explored within our group on a previous apparatus, leading to state-of-the-art two-qubit entanglement rates on the ion trap platform [4, 13], creation of highly non-classical motional states [10, 19], and a simulation of Z2 gauge theories encoded via the spin and motional degrees of freedom of a two-ion chain [saner2025real-time, 20]. However, hard limitations of both scaling these demonstrations to longer ion chains, and of interaction strength due to available optical intensity, were often restrictive. To solve both these issues, a new apparatus has been designed, constructed, and characterised, motivated by these applications. Colloquially the system is named “*FastGates*”, owing to its experimental objectives, and we shall refer to it as such throughout this chapter. These objectives give the following specific control requirements:

- Storage of medium length (5-10) qubit registers 2.2.

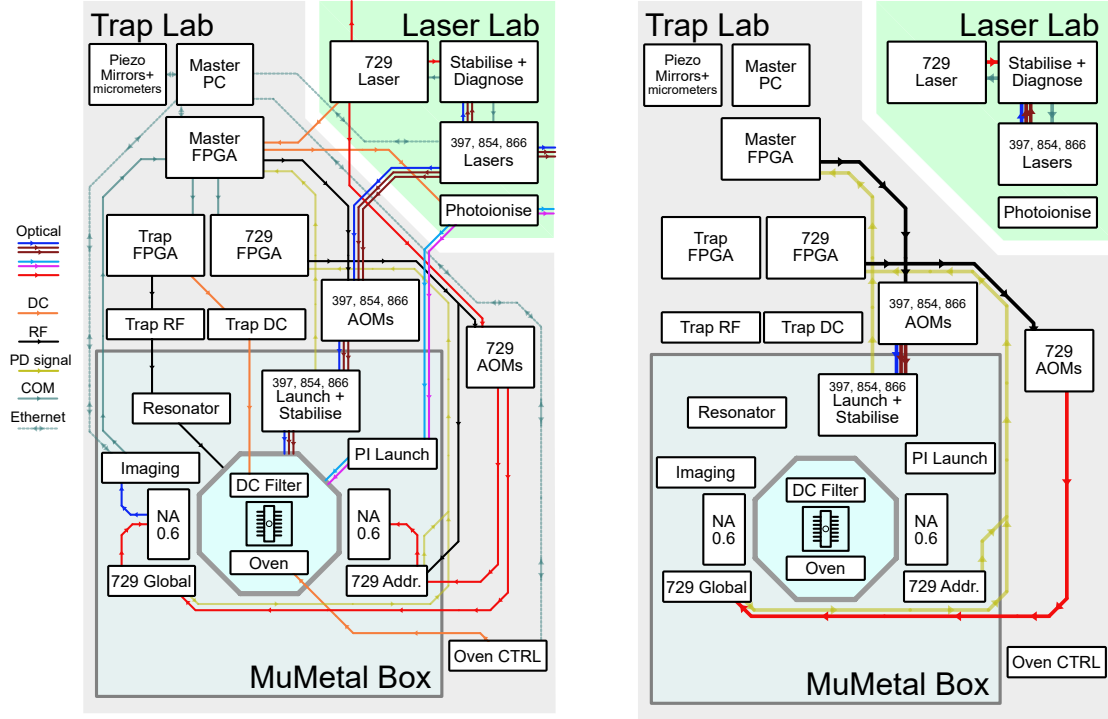


Figure 2.1: Block diagram schematic of the *FastGates* apparatus. Blocks resemble discrete subsystems, which shall be discussed within this chapter. lines resemble relevant optical (free-space or in-fibre), RF, DC, and communication paths between subsystems. The right panel highlights closed feedback loops for laser power and frequency stabilisation. Block location signifies whether the device is in-vacuo, within the magnetic shielded (MuMetal) box, within the “trap-lab”, or within the “laser-lab”.

- Individual internal state control of each qubit 2.6.4.
- Stable motional modes for storing quantum information 2.2.1.
- Dual high numerical apperture (NA) optical access for high-intensity standing waves 2.4.

As the lifetime of such an apparatus extends past a single PhD generation, this thesis serves as a point-in-time description of the system, and where applicable, the planned future extensions will be highlighted as such. An abstract schematic of the current configuration is shown in figure 2.1 with blocks resembling either electronic, mechanical, or optical subsystems.

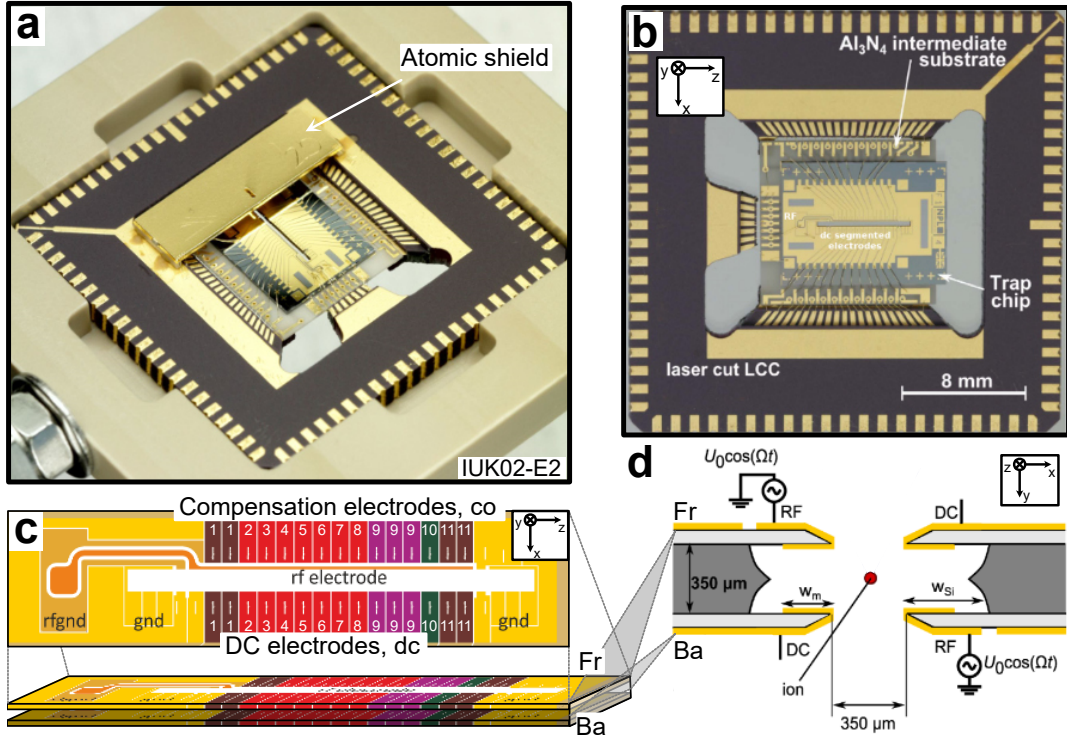


Figure 2.2: Schematics of the *NPL* trap chip used in this experiment. **a)** A photograph of the trap chip used in this experiment before being placed under vacuum. The top “atomic shield” can be seen here. **b)** A front view of the trap chip. Axial confinement is provided by a subset of the segmented DC electrodes. Radial confinement is provided by the RF rails. The trap chip and leadless chip carrier (LCC) are highlighted. **c)** A schematic of the electrode geometries and labelling for the front-face of the trap chip. A similar arrangement exists for the back-face however with the RF rail on the diagonally opposite electrode. **d)** A slice view of the trap. The red point here represents the ions, the gold are the RF and DC electrodes, and the grey is the silica substrate. It can be seen that the ion is $350 \times \sqrt{2}/2 \simeq 250 \mu\text{m}$ away from the nearest electrode. The numerical aperture of the ion plane due to the electrodes is ~ 0.71 . Figures from [21].

2.2 The Ion Trap

Recently, the micro-fabricated surface style linear Paul trap has gained popularity due to the maturity of chip fabrication technologies [22] and the potential route to scalability this offers [23]. In the surface trap, the 3D radial and axial electrodes of a “macro” trap are effectively projected onto a 2D surface. The stable confining point of such a trap is typically designed to be on the order of $50 - 100 \mu\text{m}$ above the chip surface. The maturity of chip fabrication has allowed the creation of complex multi-zone devices with many DC electrodes. These multi-zone traps enable the

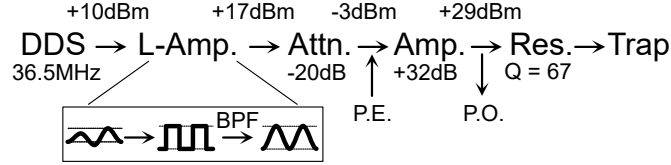
shuttling of ions, a requirement for Quantum CCD type architectures [23]. However, this surface style geometry typically comes with two costs: the depth of the trapping potential is often greatly reduced due to anharmonicity, and the close proximity of the surface to the ion can be a large contributor to motional heating rates [2, 24]. Our trap is provided by the *National Physical Laboratory* in the UK. This is a micro-fabricated 3D trap, which brings together the advantages of chip fabrication as well as the low heating rates and high trapping depths of a 3D style trap with greater ion-surface distances. Details on its design and characterisations can be found in [21, 25, 26]. Figure 2.2 shows the electrode geometry of the trap and the relevant length scales. The trap package consists of an outer leadless chip carrier (LCC) with the outer pads routing DC, RF, and ground connections to an intermediate A_3N_4 substrate. The connection to the front and back surfaces of the trap chip is made through vias and wire bonds to the intermediate substrate [wilpers2013compact]. The trap chip itself is a monolithic double-sided silicon device with gold plated electrodes on the front (Fr) and back (Ba) surfaces. These two layers are shown schematically in the bottom panel of (c). The electrode schematic for the front surface is shown, and the electrode indexing convention is highlighted. Electrodes recessed from the ion due to the RF rail are named “compensation”-electrodes (co), while electrodes on the opposite side provide the strong DC potential required for axial confinement¹. We typically trap ions centered on electrode dc5. Panel (d) shows a slice view of the trapping region, highlighting the ion position between the front and back surface electrodes. The dual RF rails and DC electrodes are shown in gold, and the undercut silicon bulk is highlighted in dark grey.

The next two subsections describe the source of RF and DC voltages to the trap electrodes. The physical mounting of the trap in the vacuum system is discussed in section 2.4, and the optical access to the trap is discussed in section ??.

¹Note: Compensation electrodes co1 and co11 on both front and back surfaces are grounded on the trap package, and so are not available for use. There is a continuity fault on front dc11 and so this electrode is also not available for use.

2.2.1 Trap RF Chain

To utilise radial motional modes for low error quantum gates, we require the radio frequency (RF) field that produces the desired trapping pseudopotential to be both frequency and amplitude stable. Furthermore, as the natural time scale for two-qubit gate durations is $\tau = 2\pi/\omega_m$ (see section ??), where ω_m is the motional mode angular frequency, then greater motional mode frequency is desired for fast entanglement. From equation 1.1, it can be seen that radial motional mode frequency is proportional to RF voltage, and inversely proportional to the frequency. Our *NPL* trap is rated for a max peak-to-peak voltage of 400 Vpp on the RF electrodes, so to target high motional mode frequencies, we want to be near this maximum value. For safe operation we aim for 360 Vpp. The RF chain which provides this signal is abstractly shown as the chain:



and embedded within the schematic 2.1. Each element of the chain is designed to provide a low-noise, high-voltage RF signal to the trap electrodes, and is discussed in turn below.

The frequency source is a direct-digital-synthesiser (DDS), named *Urukul* [kasprowicz2022urukul], part of the *Artiq Sinara* [27] hardware ecosystem. *Urukul* is used extensively in our lab for driving AOMs and here used as the source drive of our trap RF. The

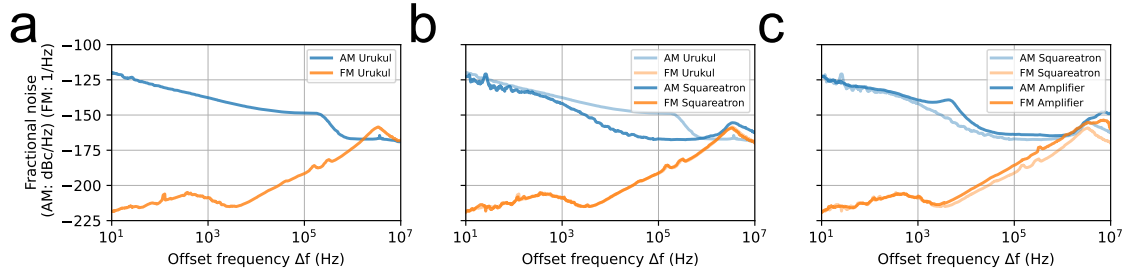


Figure 2.3: Characterisation of the RF chain. **a)** SSB spectra for AM and PM, given as fractional noise, measured after the *Urukul*. **b)** Measured after the Squareatron. **c)** Measured after the amplifier.

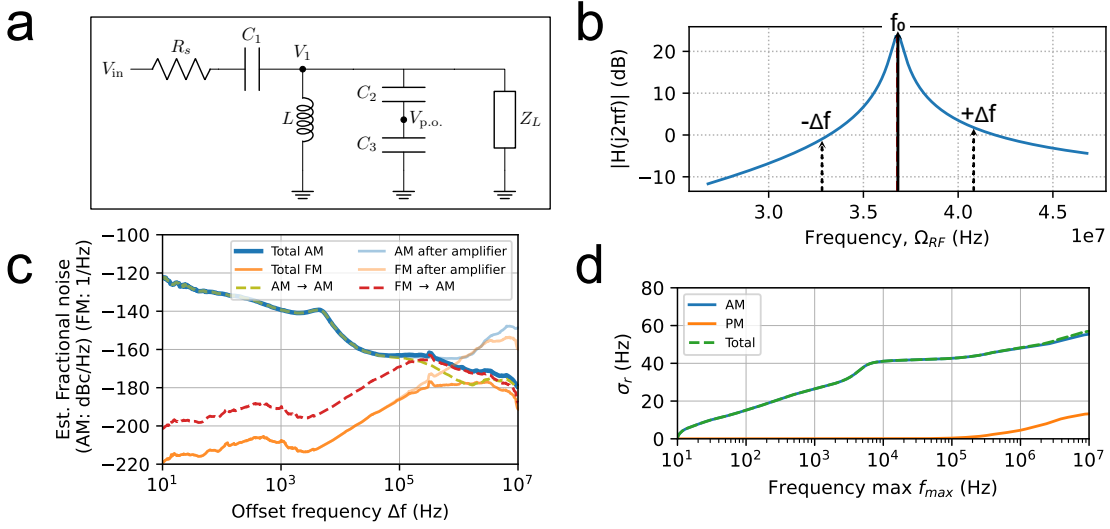


Figure 2.4: Expected SSB after the resonator, and estimated mode frequency noise. **a)** Resonator circuit diagram. **b)** Resonator lineshape, given by magnitude of the transfer function $|H(f)|$. **c)** Expected AM and FM fractional noise spectra after the resonator. **d)** Integrated RMS noise of radial motional mode frequency due to the RF chain.

Urukul is operated at maximum output power, +10 dBm, and with a frequency ~ 35 MHz (set by resonator, discussed below). As can be seen in 1.1, amplitude and frequency instability will directly lead to radial mode instability. Expanding for small amplitude, dV_{RF} , and frequency, $d\Omega_{RF}$, errors we find the motional mode frequency, ω'_r , is related to the ideal frequency, ω_r , by

$$\omega'_r = \omega_r(1 + dV_{RF}/V_{RF} - d\Omega_{RF}/\Omega_{RF}). \quad (2.1)$$

The instability of the oscillator can limit motional mode coherence time (see section 3.3.4 for characterisation with the ion). We measure noise originating from both amplitude modulation (AM) and frequency modulation (FM) across the RF chain using a phase noise analyser². This was done to verify low-noise operation of the individual devices, and to estimate expected motional mode errors due to the inherent noise on the RF chain.

We quote FM as the fractional frequency stability power spectral density (PSD),

$$S_y(f) = \frac{f^2}{f_0^2} S_\phi(f), \quad (2.2)$$

²Rhode and Schwartz FSWP PNA

where $S_\phi(f)$ is the phase noise PSD (which is the Fourier transform of the Autocorrelation function) f is the offset frequency, and f_0 is the carrier frequency. $S_y(f)$ has units of $\left[\frac{1}{\text{Hz}}\right]$. AM noise is quoted as the single sideband (SSB) power, S_α , in a 1 Hz bandwidth at an offset frequency, f , from the carrier, normalised to the total power of the carrier, and is here given in units of $\left[\frac{\text{dBc}}{\text{Hz}}\right]$.

Figure 2.3, panel (a) displays the *Urukul* AM and FM spectra. The *Urukul* is phase referenced to the master FPGA board (*Kasli*), which is in turn phase locked to an external 10 MHz atomic clock³. It can be seen that AM noise dominates up ~ 1 MHz, suggesting that this will be the limiting error source for motional mode stability. The AM noise may be reduced by using a non-linear voltage clamp. We use an “ultra-low noise limiting amplifier” named *Squareatron* developed previously in the group, and further refined by *Oxford Ionics*. *Squareatron* takes an input RF signal and converts to a stable amplitude square-wave output which is subsequently bandpassed around the desired RF frequency to recover the sinusoidal signal, but with significantly reduced AM noise. The AM and FM after the *Squareatron* is shown in panel (b). It can be seen that between 1 kHz and 1 MHz, AM is considerably suppressed. There appears to be an increase in AM at higher frequencies (ToDo: explore this further). The *Squareatron* has a 17 dBm output power, which we attenuate (by -20 dB) before an amplification stage to reach the desired power for the trap. Highlighted in the above chain schematic is the option to input a parametric excitation signal (P.E.), which we shall discuss further in section ???. For the following measurements, the P.E. branch was removed. AM and FM introduced by the low-noise amplifier⁴, can be seen in panel (c). Most notably an increase in AM is seen between 1 kHz and 10 kHz. To impedance match the 50 Ω line from the amplifier to the small capacitance of the trap electrodes and vacuum feedthrough, a resonant matching circuit is used. This “resonator” has three main effects on our system:

1. The source 50 Ω is matched to the (almost) purely capacitive load.

³ToDo: Abaqus atomic clock

⁴Mini-Circuits ZHL-1-2W-S+

2. The voltage step up provides the necessary amplitude for high motional mode frequencies.
3. The input RF is bandpassed around the desired trapping frequency.

The latter two points are practically useful, especially the bandpass effect of the resonator attenuating noise on the RF line, however, changes to the resonators Q factor and central frequency due to thermal and mechanical noise can be a significant source of low frequency motional mode instability. Opting for stability, a PCB based design⁵ with good thermal management was seen as preferable to a large helical resonator, which is often used in ion trap experiments for its high Q factor. Aiming for a moderate Q factor of ~ 50 allows for a good voltage step up, and some bandpass filtering, without being overly sensitive to thermal and mechanical noise. The circuit diagram of the constructed resonator⁶ is shown in figure 2.4. The inductor consists of XX(ToDo gauge and oxygen?) gauge Copper was made to have, $L \sim 0.8 \mu\text{H}$, with toroidal iron powder core⁷, and C0G (low tempco) rated ceramic capacitors, $C = 5.4 \text{ pF}$. The vacuum can feedthrough and trap were measured to present an $\sim 18 \text{ pF}$ load. The trap itself presents a $C_T = 4 \text{ pF}$ [wilpers2013compact], the cable between the resonator and the vacuum feedthrough contributes $\sim 4 \text{ pF}$, and the remaining presented capacitance is expected to originate from the feedthrough and in-vacuum RF line. In vacuum, the RF line is a UHV coated 0.4mm diameter single strand copper wire, which connects to an in-vacuum distribution PCB. The RF connection to the trap LCC is made by a *fuzz button* vias.

The resonator had a measured quality factor of $Q = 66.7(X)$ (ToDo: get quality factor plot), and a central frequency of $36.5(X) \text{ MHz}$. The effect of the resonator on the AM and FM noise can be calculated through the transfer function, $H(f)$, of the resonator circuit. Capacitors ($C2$ and $C3$) are used to provide a pick-off (p.o.) voltage which can be monitored, and if required, used in a closed feedback loop. We

⁵This style PCB resonator has been used around the Oxford Ion Trapping Group for some time, and is based originally on a design by Tom Harty

⁶Credit to Emily Hirsch for construction of the recent resonator.

⁷Micrometals T80-6

however have not attempted this as of yet, and shall ignore the effect of the pick-off for this analysis. The transfer function of the simplified circuit is then given by,

$$H(f) = \frac{V_{out}}{V_{in}} = \frac{Z_p}{Z_s + Z_p}, \quad (2.3)$$

where

$$\begin{aligned} Z_p &= \left(\frac{1}{j2\pi f L} + \frac{1}{Z_L} \right)^{-1}, \\ Z_s &= R_s + \frac{1}{j2\pi f C_1}. \end{aligned} \quad (2.4)$$

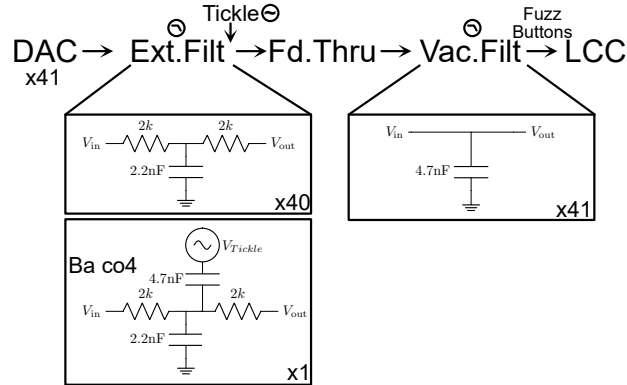
Given the approximate capacitance and inductance values of the resonator and trap, the analytic $|H(f)|$ is plotted in figure 2.4 panel (b), and shows good agreement with the measured quality factor and central frequency (here $Q = 69$ and $f_0 = 36.8\text{MHz}$). Final AM and FM spectra are calculated through considering both upper ($+\Delta f$) and lower sidebands ($-\Delta f$) following reference [danielsonAM2018]. We find that for both AM $\rightarrow$ AM and FM $\rightarrow$ FM (where " $\rightarrow$ " represents the transform from input noise type to output noise type) the input PSD are low pass filtered with a cut-off frequency of $\sim 0.25\text{ MHz}$. We also consider the cross term, FM $\rightarrow$ AM, which here we expect to be non-negligible at higher offset frequencies due to the asymmetry of the resonator lineshape. Panel (c) shows the expected fractional noise after the resonator, with total FM noise (orange line), total AM noise (blue line), and contributions from AM $\rightarrow$ AM (dashed green), FM $\rightarrow$ AM (dashed red). Integrating the AM and FM fractional noise spectra between some chosen range of offset frequencies, f_{min} and f_{max} , gives the expected fractional RMS noise on the radial mode frequency of the ions. Assuming a mode frequency of $w_r = 3\text{ MHz}$ the RMS, σ_r , is shown in figure 2.4 panel (d) as a function of f_{max} , with $f_{min} = 10\text{ Hz}$. It can be seen that the expected noise is dominated by the AM contribution, and in the frequency range considered, corresponds to a mode frequency instability of $\sigma_r = 60\text{ Hz}$. This analysis excludes any noise on the resonator itself, which may be a significant contribution at low frequencies due to thermal and mechanical noise. The motional modes will also be subject to noise from other coupling mechanisms, such as voltage noise on the DC electrodes, or fluctuating stray electric fields, which

we do not consider here.

Here we have characterised both AM and FM noise on the RF chain and estimated the expected contribution to motional mode instability. Further characterisations of the motional mode stability using the ion as a sensor is discussed in section 3.3.3.

2.2.2 Trap DC Voltages

Static fields are required for reshaping the trapping psuedo-potential. This is useful for setting desired mode frequency splittings, for shuttling ions, and for compensating stray electric fields. As shown in figure 2.2, the *NPL* trap has 40 independently controllable DC electrodes. A further DC bias voltage can be applied directly to the silicon bulk. DC Voltage is supplied to the LCC through the following chain:



and embedded within the schematic 2.1. Two multi-channel DACs, known as *Fastino*, part of the *Artiq Sinara* hardware ecosystem, provide the initial 41 DC channels. The Fastino outputs are filtered by a passive cascaded RC low pass filters with resistances, $R = 2k$, and capacitances $C_1 = 2.2$ nF and $C_2 = 4.7$ nF⁸ yielding a -3 dB “cut-off” frequency of 7.4 kHz. The low pass filter circuit is shown in the above schematic, with the first stage on an external to vacuum PCB (Ext. Filt), and the final shunt capacitor on an in-vacuum PCB (Vac. Filt)⁹ within close proximity of the trap chip to mitigate RF pickup on the DC lines. The feedthrough to the vacuum system has 51 pins¹⁰ with non-DC pins used to provide ground connections.

⁸For in-vacuum use. 1206 NPO 4.7nF 100V, SRT 1206A472JBCB

⁹Vacuum compatible PCB using Rogers RT6002. Filter boards were designed by Mariella Minder.

¹⁰*Allectra* 51-way Micro-D

On the external filter board, one channel, corresponding to electrode *Ba co4*, has a capacitively coupled in AC signal. This originates from a DDS (*Urukul*) and is used to resonantly excite the motional modes of the ion chain for characterisation and calibration purposes (see section 3.3.3). The in-vacuum PCB is connected to the trap chip via a custom PEEK interposer with embedded *Fuzz buttons* vias¹¹. The ion is confined in the “operation” zone, seen in figure 2.2 (c). The axial trapping is provided by electrodes *Ba dc5* and *Fr dc5*, where *Ba* and *Fr* correspond to the back and front plane of the trap respectively. The DAC can provide up to 10 V to these electrodes to produce an axial frequency of 1.6 MHz.

2.2.3 System Grounding

Poor ground design (or no design at all). RF signal and return in vacuum may be important.

2.3 Magnetic Field

Due to the fragility of the internal states of the ion (these are *literally* state-of-the-art magnetic field sensors), we must take great care in isolating the ion from a noisy environment. Stable Zeeman shifts of the ion energy levels are required for long spin coherence times (see section 3.2.4), and for low error single- and two-qubit gates. A permanent magnet array of Samarium Cobalt¹², $\text{Sm}_2\text{Co}_{17}$, is used in a Helmholtz configuration to create a stable magnetic field of $\simeq 5 \text{ G}$ ¹³. Samarium Cobalt was chosen for its low temperature coefficient of remenance, $-0.03\%/K$.

The ion is shielded from unwanted external magnetic fields by a custom designed MuMetal enclosure, with an internal dimension of $650 \times 650 \times 650 \text{ mm}$. The shielding consists of two layers of MuMetal shielding with internal shield thickness of 1.5mm and external shield thickness of 2.0mm¹⁴. The factory quoted shielding factor at the center of the box is 546 for DC fields. Spin coherence time comparisons with

¹¹*Custom Interconnects* Au/BeCu 0.025" diameter x 0.104" length Fuzz Button

¹²*Bunting Magnets* E643SM and E644SM.

¹³Designed and constructed by Sophie Howarth.

¹⁴*Magnetic Shields* custom design.

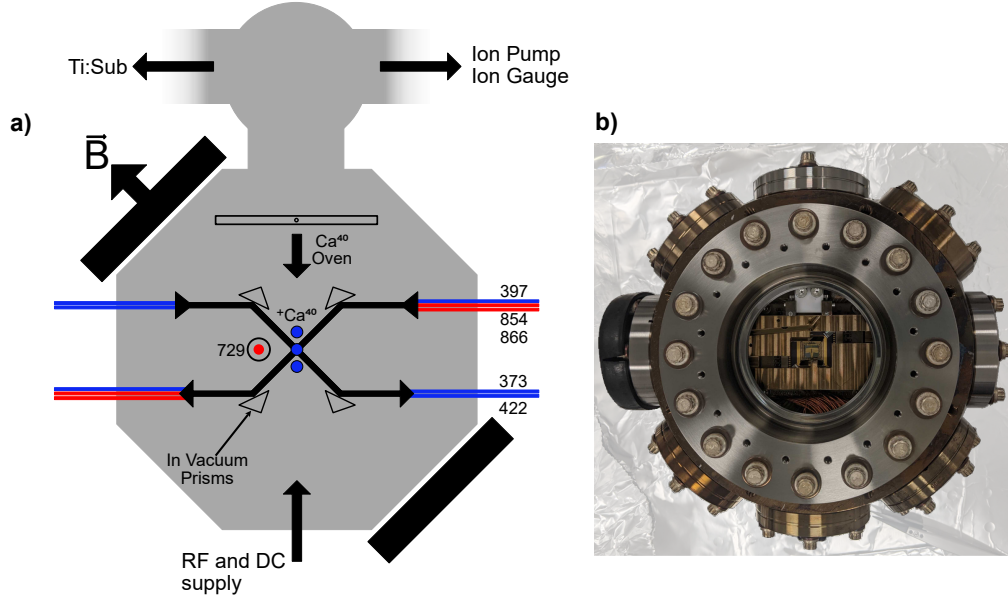


Figure 2.5: a) A schematic of the vacuum chamber. Wavelengths apart from 729-nm enter through the side CF40 viewports and are directed onto the ions by in vacuum prisms. The 729-nm light enters through the larger CF100 viewports. b) A photograph of the assembled system prior to baking.

and without magnetic shielding are shown in section 3.2.4.

2.4 Vacuum System and Beam Geometry

Ultra High Vacuum (UHV) is required to extend the ion storage lifetime. However, UHV equipment is often bulky, and puts constraints on the access and visibility of the ion chain. Here we describe the designed vacuum system, and beam access. The vacuum system and beam geometries were designed by Sebastian Saner and Mariella Minder, and constructed by Sebastian Saner, Fabian Pokorny, and myself. A residual pressure of $< 10^{-11}$ mbar is desired. For this strict requirement, care must be taken in selecting in-vacuum materials, and thorough cleaning and baking procedures must be followed. A summary of tactics that were useful in the construction of the vacuum system can be found in [28, 29].

A schematic and photograph of the vacuum system can be seen in Figure 2.5.

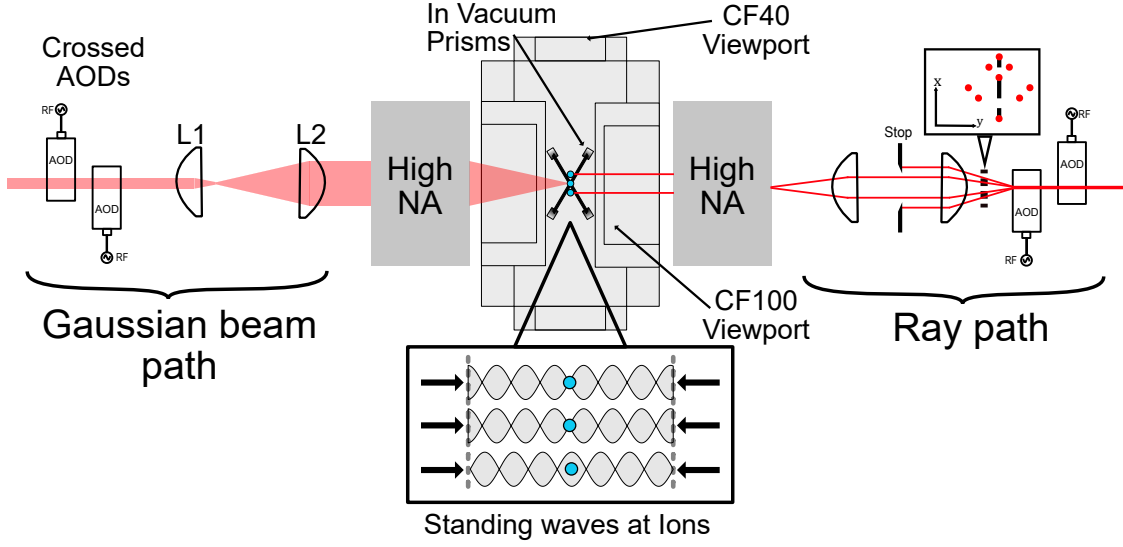


Figure 2.6: The standing wave single addressing system. Dual high NA objectives focus the light to a tight waist at the ions location. AODs are used to steer the light to only selected ions. The left hand side of the figure shows the Gaussian profile of the light, whilst the right hand side shows a ray representation of how two singly addressing spots are formed at the ions. L1 is a telecentric scanning lens and in combination with L2 form a beam expander.

The system consists of a 6" spherical-octagonal experimental chamber¹⁵ connected to a spherical chamber with ion pump¹⁶, ion gauge¹⁷ and Titanium sublimator pump (TSP)¹⁸ attached. The ion pump and TSP maintain the UHV to the desired $< 10^{-11}$ mbar. We find that on the ion pump alone UHV cannot be maintained indefinitely, however, firing the TSP every 4 weeks with 41 A for 60 seconds provides a sufficient pumping rate. At the time of system baking, a He leak test was performed, however no external leaks were found. We suspect the gradual pressure increase is either due to our use of in-vacuum optics, optics epoxy adhesive¹⁹, or our use of soldered PCB components.

For optical access, there are two recessed CF100 viewports²⁰ on the two large faces of the experiment can, see figure 2.6. Recessed viewports were required due to the

¹⁵Kimball MCF600-SphOct-F2C8

¹⁶Agilent VacIon Plus 20 Pump

¹⁷Agilent UHV-24P Ion gauge

¹⁸Scanwel custom housing

¹⁹EPO-TEK 353ND

²⁰UK Atomic Energy Authority P/N VPR100015

size of the chosen main objectives²¹. These custom objectives have an effective focal length of 33 mm, a working distance of 24.4 mm, and a numerical aperture, NA, of 0.6. They are coated for 397-nm and 729-nm. The consideration for dual high NA objectives is relatively unique in ion trap experiments, and was mainly motivated by previous work on fast entangling gates via standing waves [13].

There is further optical access via two CF40 side viewports²² coated for 397-nm, 422-nm, 729-nm, 854-nm and 866-nm, seen in figure 2.5. Due to the spatial constraints from the trap assembly and high NA objectives, multiple of our beams must enter the vacuum can through these side ports. For the ion chain to be visible from these side ports, in-vacuum dielectric mirrors are used. These fused-silica mirrors are UHV rated and coated for 372-nm, 397-nm, 422-nm, 729-nm, 854-nm and 866-nm, with reflectivities of $>99\%$ for both s- and p-polarised light. Figure 2.5 shows a schematic of the beam geometries via the side ports. A limitation of this beam geometry is that due to the permanent B-field direction, it is not possible to provide pure π or $\sigma_{\pm}$ light to the ion. However for applications where strict polarisation control is needed, the beams can be incident through the CF100 viewport.

An electrical feedthrough on a CF40 flange²³ is used to supply our trap chip and atomic source oven with DC and RF voltages. As the DC cables run within close proximity to the RF supply, electrical pick up is a potential issue for our DC lines. RF leaked onto our DC electrodes will create unwanted pseudopotential which can lead to unexpected mode geometries, or required compensation field. This leakage is mitigated through the in-vacuum RC low pass filter board within close proximity of the trap chip as mentioned in section 2.2.

²¹Photon Gear custom Atom Imager

²²LewVac ZFSVP-DUV-40CF-OUM

²³Allectra custom

2.4.1 Trap Package

2.4.2 Beam Geometry

2.5 Imaging System

The imaging system is used to collect fluorescence light from the ion chain for readout, and to image the ion chain for characterisation. We aim for a separation of 5 μm between ions, and require adequate distinguishability of ions in the imaged plane. The system consists of a high quantum-efficiency camera²⁴, an $f = 200$ mm relay lens, and the $f = 33$ mm NA=0.6 objective. This gives a magnification of 6x, and considering the pixel size of 6.5 μm , leads to a separation of 5 pixels between ions. The objective is on a 5-axis alignment stage with the z-axis focus controlled by a piezo micrometer for fine adjustment of the ion focus.

A PMT is also used...

2.6 Ca^+ Laser Systems

Singly-charged group 2 elements are popular in ion trap experiments due to their single outer electron resulting in Hydrogen-like energy levels. In this thesis we use $^{40}\text{Ca}^+$.

$^{40}\text{Ca}^+$ has no nuclear spin giving the (relatively) simple level structure shown in figure 3.1. The external magnetic field of 5.4 G is applied to split the levels via the Zeeman effect. The relevant laser transitions used in our apparatus are indicated. A zero nuclear spin isotope of calcium was chosen due to this simple level structure without hyperfine splitting. The greater number of levels due to hyperfine splitting lead to more decay paths and therefore greater complications in both cooling and gate schemes. However, due to the lack of hyperfine structure, there are no available magnetic field insensitive transitions, which are often used in ion trap experiments to further decouple the ion from a noisy environment [30].

We define our qubit by the quadrupole transition at 729-nm, which we describe in section 3.1.

²⁴Hamamatsu Orca-Fusion BT

Access to other excited levels, outside of our defined two level system, are crucial for ion trap quantum logic to enable qubit readout via state selective fluorescence (397-nm, and 866-nm transitions), and state preparation via optical pumping (397-nm, 866-nm, and 854-nm transitions). Details on these schemes are given in chapter 3. Access to two additional transitions in neutral calcium for isotope selective ionisation, 422-nm and 372-nm, is also required [31].

These transitions (apart from the 729-nm) are all driven by diode lasers²⁵, which are frequency stabilised to a reference cavity²⁶ via Pound-Drever-Hall (PDH) locking. PDH locking is used to ensure that laser frequencies are stable to <1 MHz level, well below the natural line widths of all the dipole transitions listed above. The laser frequencies are continuously monitored with a wavemeter²⁷.

Frequency and amplitude control for all lasers is achieved via acousto-optic modulators (AOMs) which are driven by Artiq integrated DDS chips, *Urukuls*. The AOMs are used in a double pass configuration to allow shifts of the laser frequency by $\pm \sim 100$ MHz, with no significant angular shifts or loss of fibre coupling.

2.6.1 Diode Lasers

2.6.2 Narrow Linewidth 729 Laser

Lasers are a key tool for creating the highly localised, strong electric field amplitudes and gradients needed to drive both carrier and sideband transitions of the trapped ion.

As shown in Figure 3.1, two sublevels within the $4S_{1/2}$ to $3D_{5/2}$ manifolds define the qubit. This is an electric quadrupole transition as $\Delta l = 2$. For the Calcium ion this transition is at 729-nm, and so a near resonance 729-nm laser is used to implement single- and multi-qubit gates (sections 3.2.3 and 3.4.2). This transition is also used after Doppler cooling for resolved sideband cooling to prepare the motional mode close to its ground state (as discussed in section 3.3.1). This transition is

²⁵All Toptica diodes. Red lasers: MDL DL pro; Blue lasers: MDL DL pro HP; 372-nm: iBEAM-SMART-375-S

²⁶Stable Laser Systems SLS-6010 4-Bore Cylindrical Cavity

²⁷HighFinesse WS7

narrow linewidth due to the long lived $3D_{5/2}$ level, and so, for power efficiency, a narrow linewidth laser must be used. For the “fast entangling gates” use-case, high intensity electric-fields are required at the ion to drive sideband transitions, this will be discussed further in the single-addressing and two-qubit gate sections, but here it is sufficient to say we require >100 mW of light at the ion plane. Here we describe the 729-nm system consisting of a Ti:Sapph laser system pumped with by an Nd:YAG 532-nm laser.

An *M2 Solstis* Ti:Sapph is pumped via 18 W of 532-nm light from a *Coherent Verdi-V* system to produce around 5W of 729-nm light. The Ti:Sapph is engineered to operate with a stable < 50 kHz linewidth. Ti:Sapph crystals have broadband gain profiles, which are often exploited in research environments to create frequency tunable laser systems. We however want a narrow linewidth, single frequency laser. The *Solstis* has multiple intracavity frequency selective elements which consist of (in order of coarse frequency selectivity), a birefringent filter, a tunable Fabry-Pérot etalon, and the surrounding bow-tie cavity. For stable single mode operation, the *Solstis* employs an active “dither” servo to lock the peak of etalon transmission to one of the cavity longitudinal mode. This dither consists of periodically varying the etalon spacing with a frequency of around 20 kHz. We must be aware of this dither frequency as the phase modulation leads to the creation of sidebands on our light which can interact with the ion causing unexpected errors in gates. This dither frequency (and harmonics of it), can be observed via composite pulse experiments on the ion, however it is currently not expected to be a limiting source of error in any of the interactions we study.

As mentioned, the *Solstis* alone can operate with linewidths of < 50 kHz, however this linewidth is pushed down further by referencing the Ti:Sapph output to an ultra high finesse cavity by *Stable Laser Systems* and applying a Pound-Drever-Hall (PDH) lock [32]. A schematic of the 729-nm system is shown in Figure 2.7. PDH locking requires applying two sidebands via an electro-optical modulator (EOM) to the light and directing it onto the stable cavity. The light reflected from the

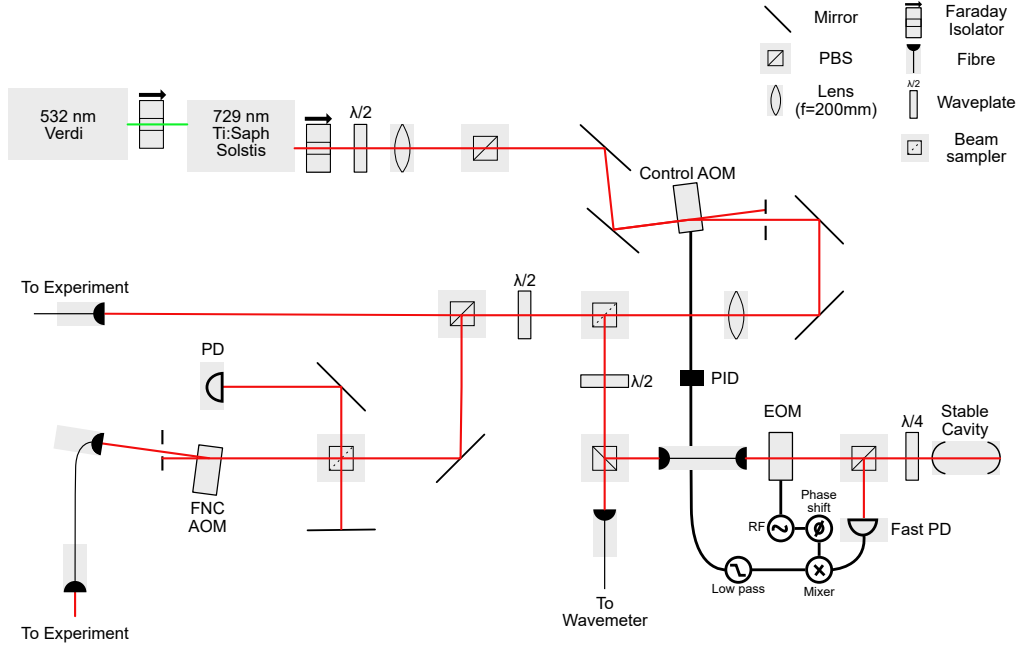


Figure 2.7: The 729-nm system. A Ti:Sapph laser tuned to 729-nm is pumped by a 532-nm source. Light is picked off at the first beam sampler to stabilise by PDH locking to a cavity.

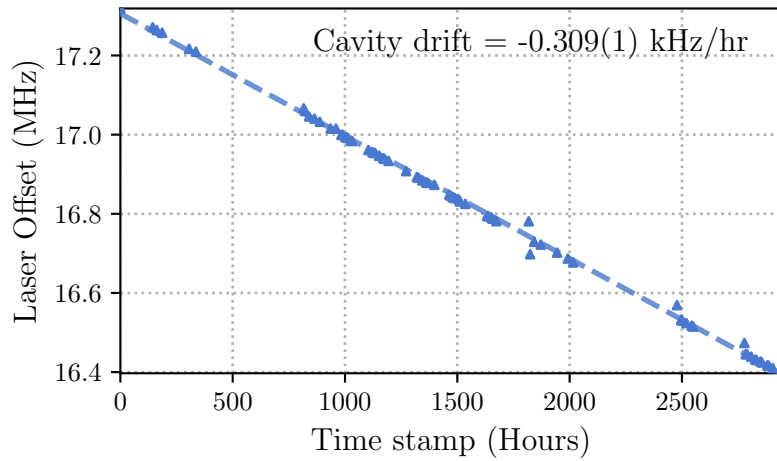


Figure 2.8: Cavity drift over 125 days measured by reference to the ion transition. The laser offset here is with respect to running all 729-nm AOMs at their central frequencies. Outliers are due to bad fits of the ion transition frequency, not due to changes in the cavity resonance.

cavity is then directed onto a fast photodetector²⁸. The reflection from the cavity consists of the interference between the carrier and the sidebands which have been respectively altered by the cavity transfer function. The photodetector signal is mixed down with the same oscillator signal as provided to the EOM but delayed by some chosen phase, and finally low pass filtered to produce a signal for use as the error signal in the servo loop. This error gives a measure for how far the carrier frequency is from the stable cavity resonant frequency and is used for feedback onto the control AOM situated after the Solstis. With this system, a linewidth of $\ll 1$ kHz for the 729-nm light is expected. The electronics for this control loop are provided also by *Stable Laser Systems* in the form of their *FPGA Servo* lock box. For an effective PDH lock, both short and long term stability of the reference cavity is required. To ensure the cavity is insensitive to the environment, it is made of an ultra low expansion material. The cavity is temperature stabilised at the zero crossing temperature of 30.6°C, and is further isolated by being housed in a vacuum system at $< 1e - 7$ mbar. Long term cavity drifts of 309(1) Hz/hr are measured over 125 days, seen in figure 2.8. This measurement uses the ion as a frequency reference to probe the cavity frequency.

Figure 2.7 displays the other beam paths for our 729-nm system. Some light is picked off and sent to a wavemeter to continuously monitor the frequency. However, the majority is coupled to two output fibres for our experiment and another within the group. The 729-nm light is transported from a dedicated laser lab to a the trap apparatus lab by a 10 m single mode polarization maintaining fibre. The fibre is beneficial in cleaning up the mode from the Ti:Sapph, however it can introduce phase noise due to mechanical and thermal effects along the 10 m length. To remove this introduced noise, passive stabilisation in the form of thick foam tubing along the fibre length, as well as active stabilisation by a fibre-noise-cancellation (FNC) technique [33], is used. We use the *Sinara Stabilizer* board, a dual channel PID microcontroller, with the *Pounder* mezzanine board, a dual channel PDH lock generator. The FNC PID software was developed by A. Agrawal. A comparison

²⁸Thorlabs PDA10A2

of spin coherence times is shown in section 3.2.4 with and without fibre noise cancellation enabled.

2.6.3 Global Addressing System

2.6.4 Single Addressing System

Single ion addressing is a requirement for executing arbitrary circuits on a register of ion-qubits. Furthermore, it is advantageous to focus the optical drive onto the ion, increasing the intensity of light, and so driving faster gates. A unique design feature of our system is the ability to produce single ion addressing standing waves whose spatial phase can be controlled individually for each ion, thus enabling any-to-any connectivity for fast entangling gates. In the future we particularly wish to extend the fast two-qubit entangling gate work demonstrated in [13] on a multi-ion chain where multiple motional modes are present and contributing to the dynamics.

The design and construction of the single addressing system was led by Iver Øvergaard and, as of writing, has been in constant use in the experiment for around 1 year. Here we shall quote the design specification and relevant details of the single addressing system. In section ?? we present the characterisation data upon installation. Further technical information can be found in IRO’s Transfer report [oevergaard2025transfer], available upon request.

Multiple methods for single ion addressing have been developed in the ion trap community, including waveguide arrays [integrated, 34], micro-mirrors [<empty citation>], spatial-light-modulators [<empty citation>], and acousto-optic deflectors [<empty citation>]. Each method has its own advantages and disadvantages in terms of speed, resolution, ease of integration, and scalability, and so choice of hardware is often a matter of experimental preference and constraints. We require:

- Addressing of up to two ions in a ~ 10 ion chain with low crosstalk to neighbouring ions.

- Fast reconfigureability of the beam position within a single experimental shot (i.e. between different gates within a single circuit execution).
- Good optical power efficiency to ensure high intensity electric fields at the ion.
- Flexibility in beam positioning to allow for fine tuning of the overlap of two counterpropagating beams used to create the standing wave.
- Individual phase control of each addressed beam.
- Beam position stability to avoid power fluctuations.
- A compact design to fit within the magnetic shielded region of the experiment.
- Good telecentricity (fixed beam k-vector direction across the scanned area) to avoid unwanted travelling wave component in a standing wave gate scheme.
- Zero frequency shift of the beam at the ion plane.

Crossed Acousto-Optic Deflectors (cAODs) are a popular choice for single ion addressing, offering fast rise times ($\sim 10 \mu\text{s}$), and good resolution over a large scanning area (~ 500 resolvable spots with 2° angular range) with zero frequency shift along the addressing axis. AODs operate through the same mechanism as AOMs, but are optimised for beam deflection rather than frequency modulation. As an addressing system cAODs have poor scaling of power efficiency with the number of addressed spots, and so are not ideal for large scale systems. As we are interested in addressing ≤ 2 ions at a time, this is not a significant issue for our system.

Figure ?? (a) shows a schematic of the constructed addressing system. Due to space constraints within the magnetic shielded region, the optical elements are located on two height levels with a periscope steering the beam between them. The beam is delivered via single-mode polarisation maintaining fibre to the upper level. The beam is steered through the first AOD, with beam angle optimised for maximum diffraction efficiency into the -1^{st} order. Two lenses in a 4f configuration re-image the deflection origin of the first AOD onto the second AOD. The second

AOD is aligned for maximum diffraction efficiency into the $+1^{\text{st}}$ order. The cAOD arrangement is termed “crossed” as the second AOD is rotated by some angle α with respect to the first, so that the deflection from the second AOD has some orthogonal component to that of the first. When the two AODs are driven with RF tones of the same frequency, the output orders are listed: (0,0), (-1, +1), (-1,0), (0,+1). The second is the desired order for single ion addressing as it has zero frequency shift yet has some overall drive-frequency dependent deflection angle. To translate this deflection angle into some physical displacement at the ion plane, a telecentric scanning lens is used. The telecentricity ensures that the beam k-vector direction is fixed across the scanned area, which is important for standing wave gate schemes to avoid unwanted travelling wave components. At the intermediate focus after the scanning lens, the unwanted orders are blocked by an adjustable slit aperture. The beam polarisation is also cleaned and selected at this stage using a polarisation filter and half-wave plate. A plate beam splitter is used within the perscope to pick off $\sim 5\%$ of the light for monitoring on a photodiode, which is used in a feedback loop to stabilise the intensity at the ion plane (the same method as discussed in section ??). The scan lens, in combination with a relay lens, forms a beam expander to increase the beam NA for the final objective lens to focus to a beam waist of $\approx 1.5 \mu\text{m}$ at the ion plane. Characterisation of the system (details given in section ??) showed that the addressing beam could be focused to a Gaussian waist of $1.50(1) \mu\text{m}$, measured by scanning a single trapped ion with Rabi experiments. The same beam was then tested in a five-ion chain, where the nearest-neighbour intensity crosstalk was measured to be $4.16(2) \times 10^{-4}$.

2.7 Control System

2.8 Ion Loading

3

Experimental Characterisation

Contents

3.1	The Ion	36
3.2	Spin	39
3.2.1	State Preparation	40
3.2.2	Measurement	41
3.2.3	Single Qubit Rotations	42
3.2.4	Spin Coherence Times	45
3.3	Motion	48
3.3.1	Cooling	48
3.3.2	Heating Rates	52
3.3.3	Motional Mode Stability	53
3.3.4	Motional Coherence Times	55
3.4	Spin-Motion	57
3.4.1	Spin-Dependent Forces	57
3.4.2	Two-Qubit Entangling Gates	61

Before we can proceed with our planned experiments involving the motion and spin of the atoms, the apparatus must be characterised. This characterisation serves two purposes: it allows us to benchmark the performance of our system against state-of-the-art results, and it reveals any current limitations that must be addressed before we can successfully carry out the experiments.

As discussed above, we are interested in two paradigms: discrete variable quantum computing (DVQC), that concerns the manipulation of qubits (or spins), and

continuous variable quantum computing (CVQC), that concerns the manipulation of qumodes (or harmonic motion).

For DVQC, we must have access to the following elementary operations:

- Qubit state preparation, here via optically pumping the electronic state of the ion into one sublevel of the $4S_{1/2}$ manifold — the lower level of our qubit, section 3.2.1.
- Measurement of the qubit state, here via state-dependent fluorescence, section 3.2.2.
- Single qubit rotations, here via interaction with the 729-nm laser, section 3.2.3
- Two qubit entangling-gates, here via spin-dependent forces (SDFs), using the 729-nm laser and the motional modes of the ion, section 3.4.2.

For CVQC, we need:

- Motional state preparation, here via Doppler cooling and sideband cooling, section 3.3.1.
- Measurement of the motional state, which is not yet implemented on our system but will follow [35].
- Single and two qumode gates, which are also not yet implemented, but will utilise SDFs following [14], section 3.4.1.

We split this chapter into interactions involving only the spin, then those focussed only on the motion, and finally we describe spin-motion interactions. First, however, we describe the ion, its level structure, and the relevant lasers to manipulate its state.

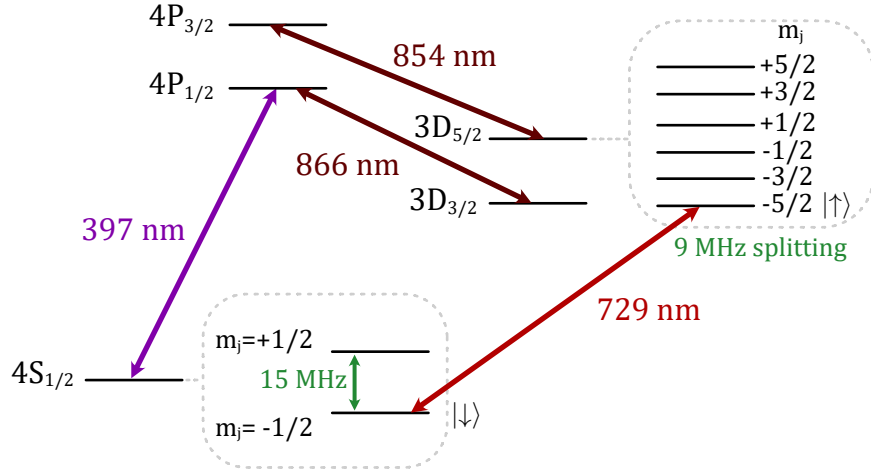


Figure 3.1: Electronic energy levels of $^{40}\text{Ca}^+$, which will be used in this thesis. The levels are split by the Zeeman effect due to a 5.4 G external magnetic field (they are shown explicitly only for the qubit manifolds). The transitions marked are required for cooling and control over the ion. The chosen qubit levels are labelled with $|\downarrow\rangle$, $|\uparrow\rangle$.

3.1 The Ion

The $^{40}\text{Ca}^+$ ion has a rich level structure, shown in figure 3.1, with many accessible quadrupole transitions $4S_{1/2} \leftrightarrow 3D_{5/2}$. The quadrupole transition is ideal for use as a qubit due to the long-lived metastable level ($\tau = 1.1$ s) [36], and the ability to optically couple the transition via a laser. As mentioned in section ??, the 729-nm beam is used for this purpose. By selecting the appropriate frequency of the 729-nm beam, qubit operations such as state preparation (section 3.2.1) and single qubit gates (section 3.2.3) may be driven, or by setting the detuning such that a motional sideband is near resonance, to couple the spin and motional degrees of freedom (section 3.4.1). Here, we discuss which quadrupole transitions are used, and the rationale behind their choice.

Zeeman splitting of the $4S_{1/2}$ and $3D_{5/2}$ states leads to 2 and 6 non-degenerate sublevels, respectively. Due to the magnetic field strength of 5.4 G (section 2.3), the S -levels are split by ~ 15 MHz, and the D -levels by ~ 9 MHz. This splitting is considerably larger than both the natural linewidth of the 729-nm transition ($\sim \text{Hz}$), and the Ti:Sapph laser linewidth (< 1 kHz), meaning the transitions can be selectively addressed by frequency tuning of the laser. There is some freedom

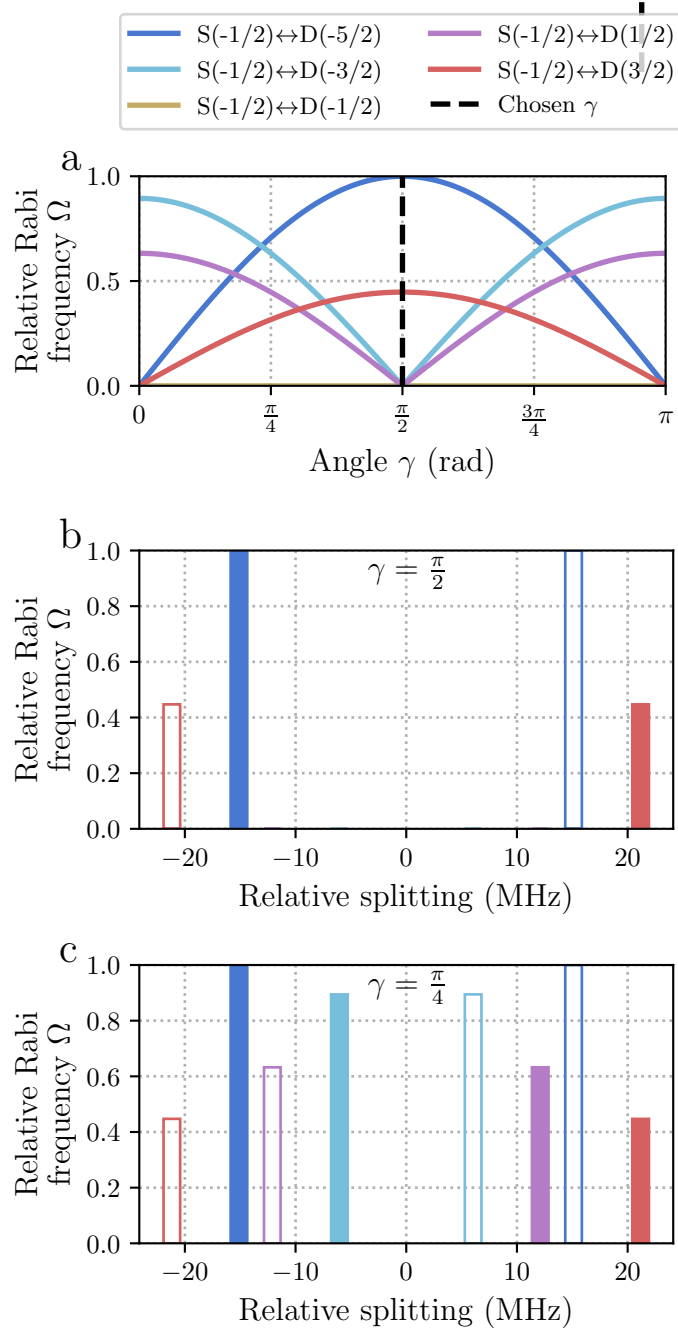


Figure 3.2: Normalised quadrupole Rabi frequencies, Ω , and frequency splittings for the $4S_{1/2} \leftrightarrow 3D_{5/2}$ Zeeman sublevels in $^{40}\text{Ca}^+$. Solid bars correspond to transitions with lower state $|4S_{1/2}, m_j = -1/2\rangle$, while open bars correspond to transitions with lower state $|4S_{1/2}, m_j = +1/2\rangle$. **a)** Relative Rabi frequencies as angle γ between the polarisation vector and the magnetic field is varied. The k-vector of the light is assumed to be perpendicular to the B-field. **b)** Relative frequency splittings and strengths when $\gamma = \pi/2$. This is the polarisation chosen for the current and future experiments to reduce the number of transitions which can cause off-resonant interactions when the qubit defined by $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -5/2\rangle$ is used. **c)** Relative frequency splittings and strengths when $\gamma = \pi/4$. At this angle, all transitions are accessible apart from those with $\Delta m_j = 0$. This is the polarisation chosen for our initial experiments to allow access to all transitions for characterisation.

in choosing which of these 10 possible (with $\Delta m_j \leq 2$) quadrupole transitions we define as our qubit. Each transition has a varying coupling strength, here denoted by the relative Rabi frequency, Ω , depending on the polarisation and k -vector of the interacting light with respect to the B-field axis. The desired criteria for qubit and polarisation choice are as follows:

- Access to a closed-cycle transition for sideband cooling, (for details on this, see section 3.3.1). Closed-cycle here means $|4S_{1/2}, m_j = -1/2\rangle \rightarrow |3D_{5/2}\rangle \rightarrow |4P_{3/2}, m_j = -3/2\rangle \rightarrow |4S_{1/2}, m_j = -1/2\rangle$.
- The qubit transition needs a large quadrupole matrix element, leading to a high Rabi frequency, Ω , to enable laser-power-efficient, fast, gates. Polarisation, γ , may be optimised to increase this coupling.
- Ideally the qubit transition should have a low magnetic field sensitivity, to reduce the effect of magnetic field noise. Using a MuMetal enclosure around the experiment does relax this constraint, see section 2.3.
- To reduce the effects of off-resonant interactions due to nearby transitions, it is desirable to limit the number of accessible transitions near to the qubit transition. This is done by selecting a polarisation, γ , such that unwanted transitions are suppressed.

Given these criteria, we use the transition $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -5/2\rangle$ with magnetic field sensitivity $-0.446 \text{ MHz G}^{-1}$. Closed-cycle sideband cooling is possible with this same transition, and state preparation via optical pumping is done via $|4S_{1/2}, m_j = +1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -3/2\rangle$ with magnetic field sensitivity $-0.624 \text{ MHz G}^{-1}$.

Figure 3.2 *a*, shows the normalised Rabi frequencies of each transition as the angle γ between polarisation vector and B-field is varied. Here only linearly-polarised light with k -vector perpendicular to the B-field direction is considered. The polarisation of the 729-nm laser is set to $\gamma = \pi/2$ (black dashed line), which maximises the Rabi frequency of the qubit transition, while suppressing nearby

Parameter	Value
729-nm laser power	1000 μW
729-nm laser detuning	0 MHz*
729-nm pulse duration	5 μs
854-nm laser power	60 μW , $250 \times I_{\text{SAT}}$
854-nm laser detuning	0 MHz
854-nm pulse duration	40 μs
866-nm laser power	15 μW , $\sim 70 \times I_{\text{SAT}}$
866-nm laser detuning	0 MHz
866-nm pulse duration	30 μs

Table 3.1: * With respect to qubit transition frequency.

Experimental parameters used for state preparation of the ion into the $|\downarrow\rangle = |4S_{1/2}, m_j = -1/2\rangle$ state.

transitions apart from that needed for state preparation. Figure 3.2 *b*, shows the relevant frequency splitting of the transitions at $\gamma = \pi/2$, which is the set polarisation for all experiments apart from initial quadrupole characterisation. For initial characterisation of the quadrupole transitions, the polarisation was set to $\gamma = \pi/4$, which allowed access to all transitions apart from those forbidden by the beam geometry, $\Delta m_j = 0$, see figure 3.2 *c*.

Other required lasers for manipulating the ion are: the 397-nm and 866-nm lasers used to form a closed-cycle for fluorescence, the 854-nm laser used to deshelve the ion from the $3D_{5/2}$ level for qubit state preparation. Details on these lasers can be found in section 2.6.

3.2 Spin

Discrete variable quantum computing consists of manipulating many two-level systems, which we refer to either as spins, or qubits. In this section the methods used to manipulate these spins are described, and their performances are benchmarked.

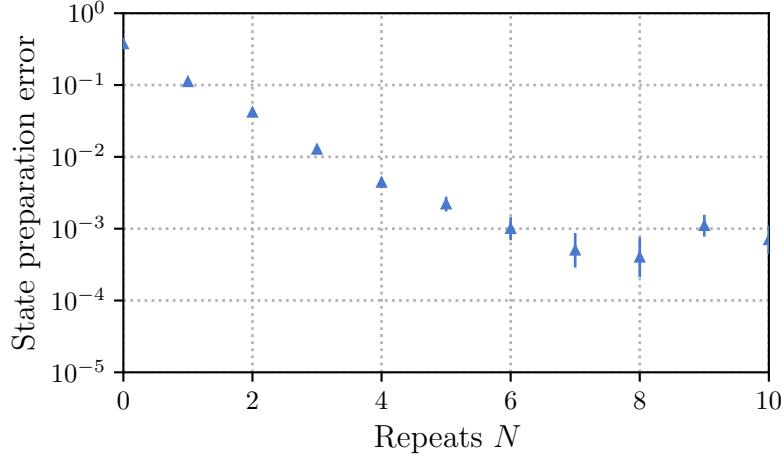


Figure 3.3: State preparation error as a function of the number of state preparation pulses, N . The error is defined as population not in the $|\downarrow\rangle$ state after the preparation sequence has been applied. The error is reduced to below 10^{-3} after $N = 7$ pulses, which is used in our experiments.

3.2.1 State Preparation

To utilise two levels of the ion as a qubit, the electronic state must first be prepared into the qubit manifold. Here, the ion is prepared into the $|\downarrow\rangle = |4S_{1/2}, m_j = -1/2\rangle$ state. This is done via effective optical pumping by pulsing the quadrupole transition $|4S_{1/2}, m_j = +1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -3/2\rangle$, followed by deshelling pulses using the 854-nm, 866-nm lasers on resonance. The 729-nm laser pulse duration and intensity is set to invert the populations, i.e. perform a π -pulse. Details for laser parameters are summarised in table 3.1. These three pulses are repeated N times sequentially. Figure 3.3 shows the state preparation error with increasing N . The error is found by applying a π -pulse on the state preparation transition directly after the state preparation sequence and looking at the population in the $|\downarrow\rangle$ state. Any reduction from $|\downarrow\rangle = 1$ indicates residual population in the $|4S_{1/2}, m_j = +1/2\rangle$ state, or the D levels. After $N = 7$ pulses, which is used in our experiments, there is a state preparation error of $< 10^{-3}$. This is further verified by randomised benchmarking in section 3.2.3. The state preparation error is likely limited by accidental excitation of the $|4S_{1/2}, m_j = -1/2\rangle$ state to the $D_{5/2}$ level. This may

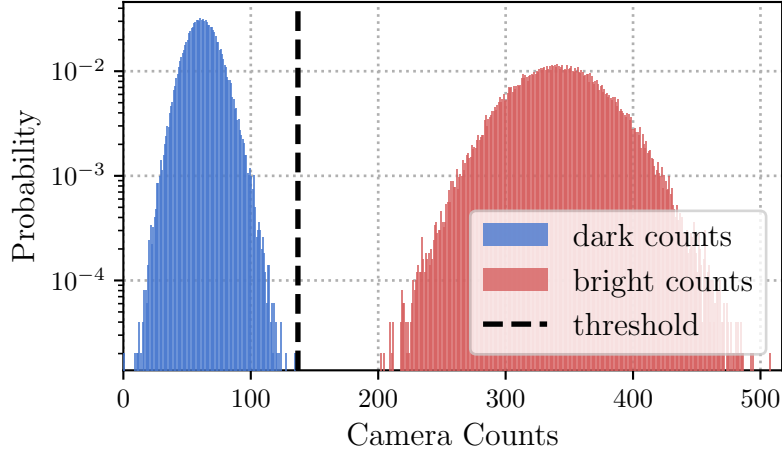


Figure 3.4: Readout histogram with 100,000 individual measurements for bright (red) and dark (blue) populations. The camera counts are presented as a histogram and fitted to Poissonian distributions to calibrate a threshold for discerning if the ion is bright or dark with the lowest probability for false measurement. Dark counts are mainly due to 397-nm light scattered from the nearest trap electrodes. With 100 μ s readout duration, and laser parameters set to Doppler cooling parameters, a threshold count of 137 photons is found, giving an expected statistical error of 6×10^{-9} given the fitted Poissonians.

be due to off-resonant excitation of another transition, or due to spectral purity of the 729-nm laser.

Another protocol for state preparation is optical pumping on the dipole transition (397-nm), which often is much faster. However, in this setup it is impractical as polarisation selectivity of the 397-nm transition is not possible due to the constraint of beam geometry from the in vacuum optics, see section 2.4, and frequency selectivity is not possible due to the low magnitude fixed B-field providing Zeeman shifts less than the natural linewidth of the 397-nm transition, see section 2.3.

3.2.2 Measurement

To measure the qubit state, the 397-nm and 866-nm lasers are applied, and the scattered 397-nm photons are imaged by the objective onto a camera (see section 2.5). 397-nm light will only be scattered if the qubit is in $|\downarrow\rangle$. Use of a high numerical aperture ($\text{NA} = 0.6$) objective greatly increases the collection efficiency. This efficiency allows the 397-nm beam to be red-detuned to Doppler cooling parameters

Parameter	Value
397-nm laser power	3 μW , $1 \times I_{\text{SAT}}$
397-nm laser detuning	-17.5 MHz
866-nm laser power	15 μW , $\sim 70 \times I_{\text{SAT}}$
866-nm laser detuning	0 MHz
Doppler cooling duration	1000 μs
Readout duration	100 μs
Camera threshold	137 counts

Table 3.2: Experimental parameters used for Doppler cooling and state-selective fluorescence of the ion.

(see section 3.3.1), while still maintaining low readout duration (100 μs) and low readout error. Cooling during readout increases the experiment's repetition rate by reducing the amount of dedicated cooling time required.

The parameters used for readout are summarised in table 3.2, and a typical histogram of readout counts for one ion can be seen in figure 3.4. The readout threshold is calibrated by taking bright counts with the 397-nm and 866-nm lasers on, and dark counts with the 397-nm laser on and the 866-nm laser off. This assumes dark counts are due to 397-nm scattering from the trap electrodes. Fitting Poissonian distributions to the bright and dark measurements, an optimal count threshold of 137 is found, shown by black dashed line. This threshold gives an expected statistical readout error of $\epsilon_{\downarrow} = \epsilon_{\uparrow} = 6 \times 10^{-9}$, where $\epsilon_{\downarrow, \uparrow}$ are the expected error associated with measuring bright or dark. This measurement ignores errors due to the finite lifetime of the metastable $3D_{5/2}$ level, which dominate $\epsilon_{\uparrow}$. The expected error due to the finite lifetime, τ , is $\epsilon_{\text{decay}} = 1 - e^{-t/\tau}$, where t is the readout duration. For $^{40}\text{Ca}^+$, $\tau \sim 1.1$ s, $\epsilon_{\uparrow} \approx \epsilon_{\text{decay}} = 10^{-4}$. Assuming the metastable lifetime is limited only by spontaneous decay overlooks leakage from the 854-nm deshelving beam, which could significantly shorten τ [34].

3.2.3 Single Qubit Rotations

A qubit is fully described by $|\psi\rangle = \alpha|\downarrow\rangle + \beta|\uparrow\rangle$, where α and β are complex numbers satisfying $|\alpha|^2 + |\beta|^2 = 1$. The qubit state can be manipulated by applying single

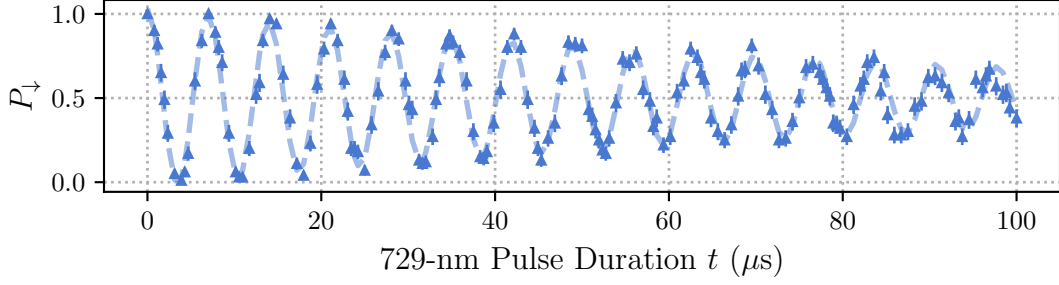


Figure 3.5: Long duration, t , Rabi flop on the qubit transition to extract Rabi frequency. Estimated probability of being in $|\downarrow\rangle$, $P_{\downarrow}$, at each time step using 200 individual experimental shots. Fitted dotted line using decaying oscillation model. Extracted Rabi frequency $\Omega/2\pi = 0.2864(2)$ MHz, and decay rate $\lambda = 0.0107(7) \mu\text{s}^{-1}$.

qubit rotations, $R_{x,y}(\theta)$, commonly referred to as gates. In the Bloch sphere, x - and y -rotations are performed by perturbations resonant with the qubit transition, here, using the 729-nm laser. The rotation angle, θ , is given by $\theta = \Omega t/2$, where Ω , is the Rabi frequency, and t is the pulse duration. The rotation axis, is determined by the relative phase of the light field and the qubit. We define the initial relative phase to correspond to a y -rotation.

Gate Time Calibration

The pulse duration, for a $R(\pi/2)$ ($\pi/2$ -pulse), is calibrated by Rabi oscillations. Figure 3.5 shows the population in $|\downarrow\rangle$, $P_{\downarrow}$, calculated from 200 experimental shots. To find the Rabi frequency, Ω , the data is fitted to a decaying oscillation model,

$$P_{\downarrow} = \frac{1 + e^{-\lambda t} \cos(\Omega t/2)}{2}, \quad (3.1)$$

where λ is the decay rate due to decoherence effects [2]. The Rabi frequency is found to be $\Omega/2\pi = 0.2864(2)$ MHz, and the decay rate $\lambda = 0.0107(7) 1/\mu\text{s}$. From this $R(\pi/2) = 1.7 \mu\text{s}$ can be estimated. These measurements were taken with 1 mW of 729-nm power, with beam radius $\sim 15 \mu\text{m}$ at the ion.

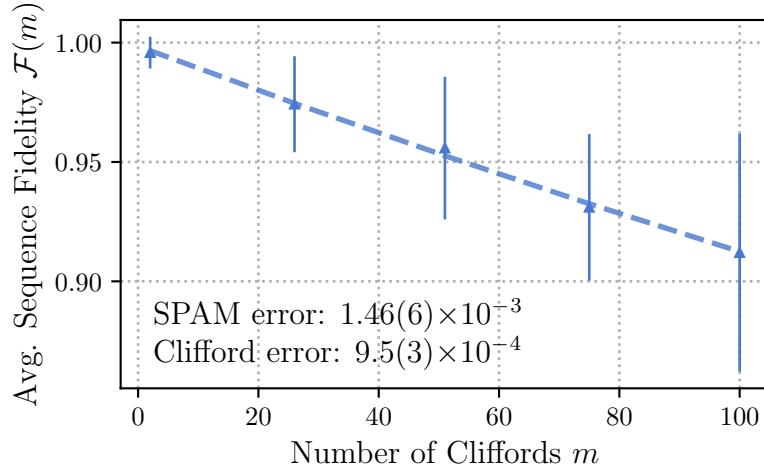


Figure 3.6: Estimating single qubit Clifford fidelities, $\mathcal{F}$, and errors due to state preparation and measurement, using randomised benchmarking. The data points are the average survival population after applying a sequence of m random Clifford gates, and a final inverting Clifford gate. The dashed line is the fit to the decay model given by equation 3.2. The error bars are given by the standard deviation of the survival populations.

Randomised Benchmarking

To evaluate the quality of both our state preparation and single qubit rotations, randomised benchmarking (RBM) [37] is employed. RBM consists of applying random combinations of gates from the single-qubit Clifford group to estimate an average error per gate. The single-qubit Clifford group is the set of unitaries which map the Pauli matrices to one another through conjugation. The RBM protocol used here is described in [38]. The sequence of operations is as follows:

1. Prepare the qubit in $|\downarrow\rangle$ via optical pumping, section 3.2.1.
2. Apply a random sequence of m single qubit Clifford gates, where m is the length of the sequence.
3. Apply a final “inverting” Clifford gate, which is chosen such that the full sequence performs the identity operation.
4. Measure the population in $|\downarrow\rangle$, $P_{\downarrow}$, via state-selective fluorescence, section 3.2.2.

5. Repeat steps 1-4 multiple times to find the average survival population, $\langle P_{\downarrow} \rangle$.
6. Repeat steps 1-5 for many different random sequences with a range of sequence lengths, m .
7. Fit the decay of the average survival population, $\mathcal{F}(m) = \langle P_{\downarrow}(m) \rangle$, to a decay model to find the average error per Clifford gate.

The decay model used to find the fidelity versus number of Clifford gates is given by [38],

$$\mathcal{F}(m) = \frac{1}{2} (1 + (1 - 2\epsilon_{\text{SPAM}})(1 - 2\epsilon_c)^m), \quad (3.2)$$

where $\mathcal{F}(m)$ is the fidelity of the sequence of length m , ϵ_{SPAM} is the state preparation and measurement error, and ϵ_c is the average error per Clifford gate. The Clifford gates are decomposed into sequences of $\pi/2$ - and π -pulses about either the x - or y -axes. Up to $m = 100$ Clifford gates are probed, and the decay of the fidelity, $\mathcal{F}(m)$, is fit using the above model. The error per Clifford is found to be $\epsilon_c = 9.5(3) \times 10^{-4}$, while the SPAM error is $\epsilon_{\text{SPAM}} = 1.46(6) \times 10^{-3}$. The decay plot for this RBM sequence can be seen in figure 3.6. The independently characterised state preparation (section 3.2.1) and measurement (section 3.2.2) errors of $\sim 10^{-3}$, and $\sim 10^{-4}$ respectively, are consistent with the measured ϵ_{SPAM} .

3.2.4 Spin Coherence Times

Decoherence, as can be seen in figure 3.5, can be caused by laser frequency and power fluctuations, magnetic field noise, and spontaneous emission from the $3D_{5/2}$ level. The coherence time of the qubit is a useful diagnostic for finding unwanted noise sources, and for benchmarking the performance of the qubit as a store of quantum information.

To measure the coherence times, we perform Ramsey sequences with varying delay durations. The Ramsey sequence is as follows:

1. Prepare the qubit in $|\downarrow\rangle$ via optical pumping, section 3.2.1.

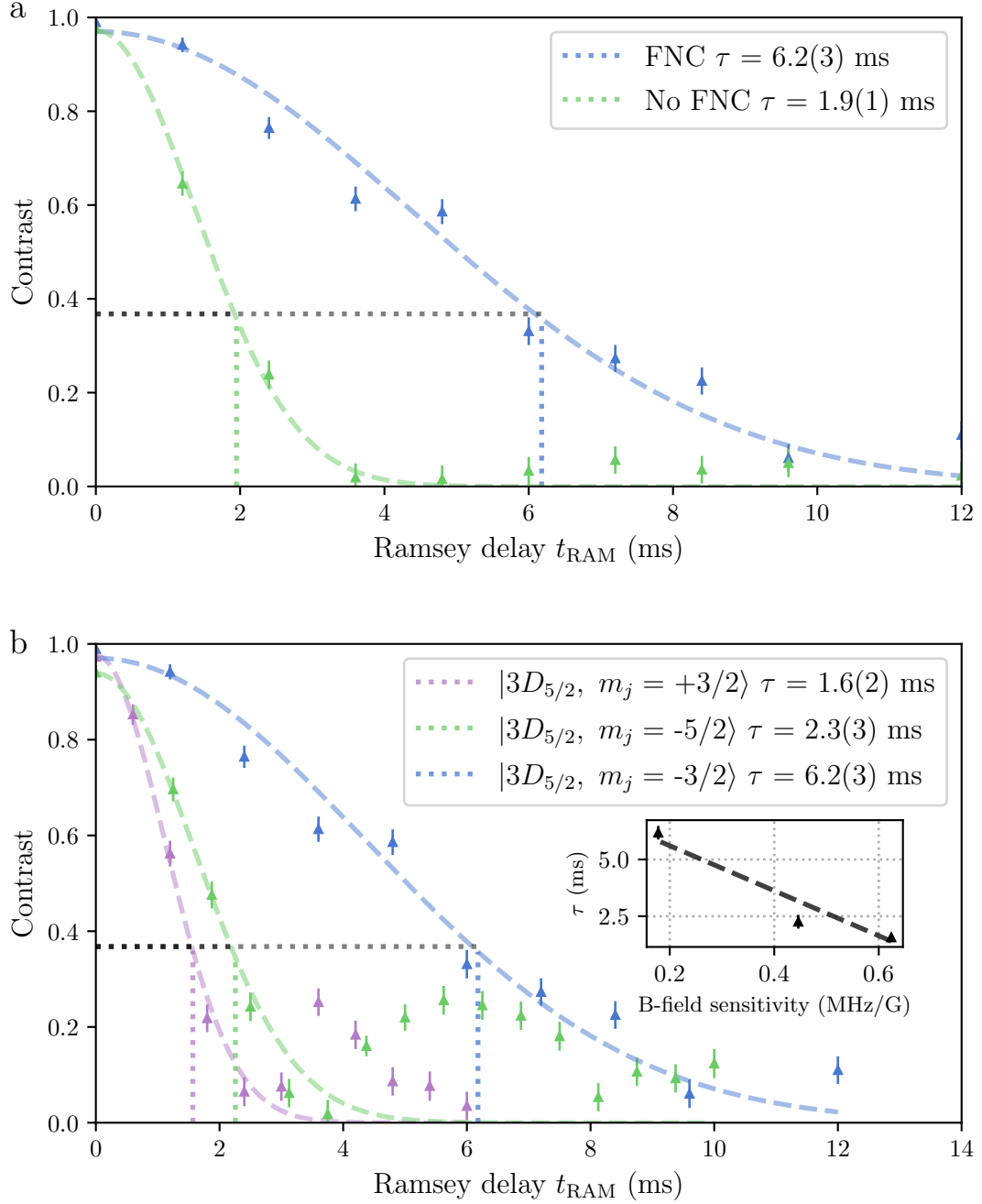


Figure 3.7: Coherence time measurements. **a)** Coherence time of the $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -3/2\rangle$ transition with and without fibre noise cancellation (FNC). **b)** Coherence time of three quadrupole transitions with varying magnetic field sensitivities. The inset plot shows a linear fit between the coherence time and the magnetic field sensitivity of the transition.

2. Apply a $R_y(\pi/2)$ pulse to prepare the qubit in $|+\rangle = \frac{1}{\sqrt{2}}(|\downarrow\rangle + |\uparrow\rangle)$ superposition state.
3. Wait for a time t_{RAM} .
4. Apply a second $R_\phi(\pi/2)$ -pulse, with a variable axis of rotation, ϕ .
5. Measure the population in $|\downarrow\rangle$, $P_\downarrow$, via state-selective fluorescence, section 3.2.2.
6. Repeat steps 1-5 for different values of ϕ and fit the resulting fringe to find the contrast.
7. Repeat steps 1-6 for different values of t_{RAM} to find the decay in contrast with time. The coherence time is defined as the time at which the contrast has decayed to $1/e$ of its initial value.

With this tool, the efficacy of the fibre noise cancellation (FNC, see section ??) can be measured. Figure 3.7 *a* shows the coherence time of the $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -3/2\rangle$ transition with and without FNC. The coherence time is increased from 1.9(1) ms to 6.2(3) ms with FNC, showing that without FNC, the coherence times are limited by laser phase noise.

With FNC enabled, to see if the coherence time is now limited by magnetic field noise, transitions with varying magnetic field sensitivities are compared. Figure 3.7 *b* shows the coherence time of three quadrupole transitions with varying $D_{5/2}$, m_j numbers: $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = \{-5/2, -3/2, +3/2\}\rangle$. These transitions have magnetic field sensitivities -0.446 , -0.178 , $+0.624$ MHz G⁻¹ respectively. The contrast decays for all three qubits are shown in the figure. The inset plot shows a well correlated linear fit between the coherence time and the magnetic field sensitivity, suggesting that the coherence time is limited by magnetic field noise. For both $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -5/2\rangle$ and $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = +3/2\rangle$, revivals in contrast can be seen which suggest that the dominant noise source is oscillatory and coherent over the Ramsey scan duration.

These measurements were taken without the MuMetal enclosure (section 2.3) fully

assembled around the vacuum system, which we expect will greatly reduce external magnetic field noise, and therefore increase the coherence times.

3.3 Motion

Motion of ions trapped in the same potential can be described by a set of normal modes [39]. These modes are labelled with respect to the ion chain geometry: axial modes oscillate parallel with the chain, and radial modes oscillate perpendicular to the chain. Due to the geometry of the 729-nm beam access with respect to the ion chain (see section 2.4), only the radial modes can be addressed in our system. For the *NPL* trap (see section 2.2), the radial modes frequencies are non-degenerate. In our system, for a single ion, the radial mode frequencies are $\omega_{\text{upper}}/2\pi = 4.0$ MHz, and $\omega_{\text{lower}}/2\pi = 2.5$ MHz, and the axial mode frequency is $\omega_{\text{axial}}/2\pi = 1.6$ MHz. All work here is using the upper radial mode.

The motional mode, i , leads to sidebands in the 729-nm spectrum at $\omega_0 \pm \omega_i$ [40], where ω_0 , the carrier, is the frequency of the 729-nm laser when on resonance with the qubit transition. The $+\omega_i$ sideband is referred to as the blue sideband (BSB), and the $-\omega_i$ is the red sideband (RSB). The RSB, BSB, and RF fields near to resonance with the motional modes, are the tools used in this section to manipulate the motion of the ion.

The motional state preparation, heating rate, frequency stabilities, and coherence times are characterised in this section.

3.3.1 Cooling

For any interaction that involves the motion of the ion, it is essential to both prepare and measure the motional state with high fidelity. For all motional interactions considered in this thesis, the ion is initially prepared as close as possible to the motional ground state, with a mean phonon number $\bar{n} \approx 0$. This is done by an initial stage of Doppler cooling, followed by pulsed sideband cooling on the 729-nm transition. The final motional state is characterised by measuring the population in

Fock states using RSB and effective BSB pulses.

Doppler Cooling

Doppler cooling relies on the fact that light incident on a moving ion appears frequency-shifted in the ion's rest frame. For Doppler cooling of $^{40}\text{Ca}^+$, both the 397-nm and 866-nm lasers are applied. The 397-nm laser is red-detuned from the dipole $|4S_{1/2}\rangle \leftrightarrow |4P_{1/2}\rangle$ transition, leading to preferential scattering of 397-nm photons that have a net cooling effect [41]. The excited $4P_{1/2}$ level decays via two channels: emission of a 397-nm photon back to $|4S_{1/2}\rangle$, or an 866-nm photon to $|3D_{3/2}\rangle$, with a branching ratio of 14.5 [42]. Since many photon scattering events are required, the cooling cycle must be closed. To prevent population trapping in the metastable $|3D_{3/2}\rangle$ level, the 866-nm laser is applied on resonance during cooling.

Equilibrium is reached when the cooling rate balances photon recoil heating. Assuming the 397-nm transition is broadened only by its natural linewidth γ , the minimum Doppler temperature is,

$$T_{\text{Doppler}} \approx \frac{\hbar\gamma}{2k_B}, \quad (3.3)$$

where $\hbar$ is the reduced Planck constant, and k_B is Boltzmann's constant [41].

For $^{40}\text{Ca}^+$, the natural linewidth of the 397-nm transition is $\frac{\gamma}{2\pi} = 21$ MHz[43], giving $T_{\text{Doppler}} = 0.5$ mK. For a radial mode frequency of $\frac{\omega}{2\pi} = 4$ MHz, the expected mean phonon number is,

$$\bar{n} = \frac{1}{e^{\hbar\omega/k_B T} - 1}, \quad (3.4)$$

yielding $\bar{n} = 2.3$ at the Doppler limit. The parameters used here for Doppler cooling are summarised in table 3.2.

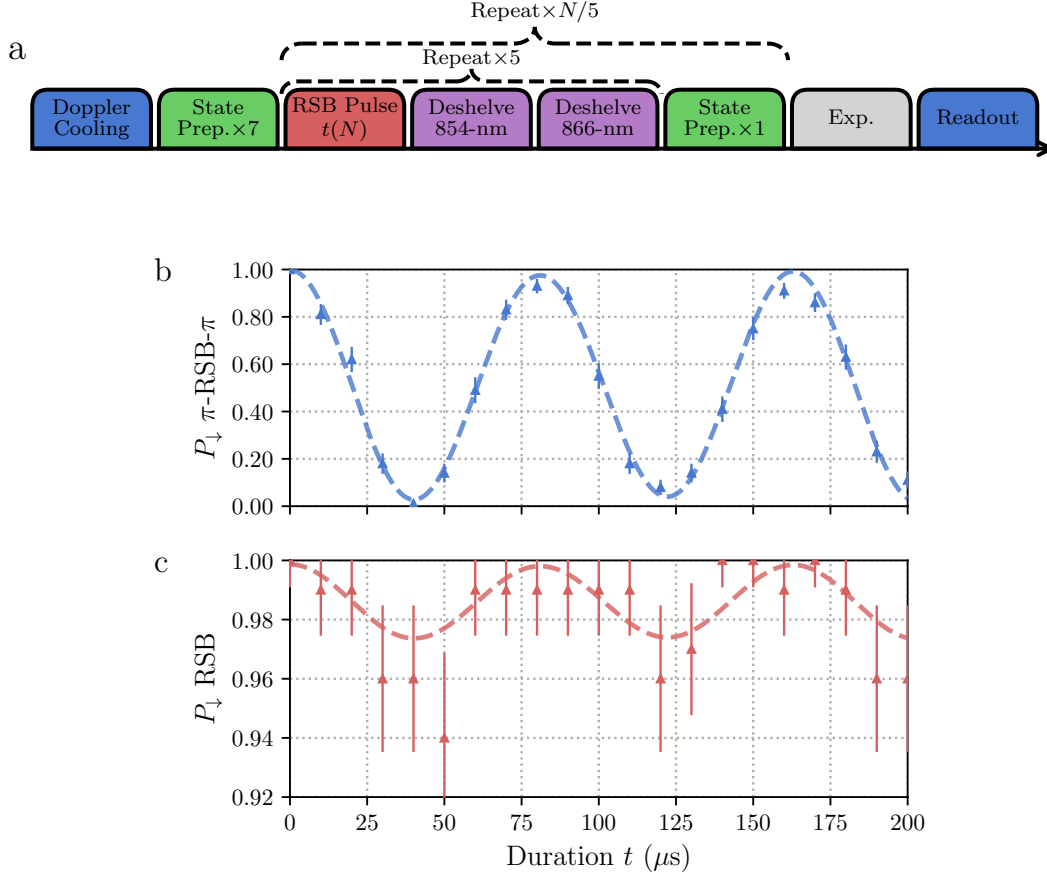


Figure 3.8: **a)** Example pulse sequence for one experimental shot including pulsed sideband cooling. The *Exp.* block represents an arbitrary sequence of operations, for the experiment under study, such as a thermometry sequence. The blue blocks represent Doppler cooling and readout (section 3.2.2), the green blocks represent qubit state preparation (section 3.2.1), the red blocks are RSB pulses with varying durations $t(N)$, and the purple blocks are deshelling pulses. After every 5 sideband cooling pulses, there is an interleaved state preparation sequence to ensure there is no population trapped outside of the closed cooling cycle. The RSB length is linearly increased with pulse number N , upto the expected RSB $R(\pi)$ -pulse duration for $n = 1$. **b, c)** Thermometry scans after Doppler and sideband cooling. The red points are the populations measured via on-resonance RSB pulses, and the blue points are the populations after a sequence of $R(\pi)$ -RSB- $R(\pi)$ with varying RSB probe durations. The dashed lines are fits to a thermal Fock state distribution, with truncation at $n = 100$. The extracted mean occupation number is $\bar{n} = 0.03(1)$, and $\eta\Omega/(2\pi) = 11.6(1)$ kHz.

Parameter	Value
729-nm laser power	3.0 mW
729-nm laser detuning	-4.0 MHz*
729-nm pulse duration $N = 1$	2.5 μ s
729-nm pulse duration $N = N_{\max}$	33 μ s
Repumping 866-nm duration	14 μ s
Deshelving 854-nm duration	5 μ s
Number of cooling cycles $N_{\max}$	50

Table 3.3: *729-nm detuning is relative to the $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -5/2\rangle$ transition frequency.

Experimental parameters used for sideband cooling. Other parameters for the repump and deshelve pulses are the same as for state preparation, see table 3.1. The 729-nm laser is detuned to the RSB resonance for the upper radial mode, $\omega_{\text{upper}}/2\pi = 4.0$ MHz, and the pulse duration for $N = N_{\max}$ is chosen to be the expected RSB $R(\pi)$ -pulse duration for $n = 1$.

Sideband Cooling

To further cool the ions toward their motional ground state, resolved sideband cooling is used. As mentioned above, the motion of the ion modulates the absorption spectrum of the ion, leading to red and blue sidebands (RSB and BSB). For the $|4S_{1/2}\rangle \leftrightarrow |3D_{5/2}\rangle$ transition, at appropriate motional mode frequencies, interaction strengths, and using a narrow linewidth laser (section ??), these sidebands can be resolved spectroscopically. The pulsed sideband technique employed consists of RSB pulses on the 729-nm transition, followed by deshelving, and repumping pulses on the 854-nm and 866-nm transitions respectively. For efficient cooling, initial short 729-nm pulses are applied to preferentially excite the thermal population with high phonon number, as the pulse duration for RSB $R(\pi)$ is proportional to $\sqrt{n}$, where n is the phonon number. The RSB pulse length is linearly increased to the expected RSB $R(\pi)$ -pulse duration for $n = 1$.

An example pulse sequence for a single experimental shot using sideband cooling, can be seen in figure 3.8 *a*. Experimental parameters used are summarised in table 3.3.

To verify the efficacy of our sideband cooling, thermometry experiments are performed by probing Fock state populations. This is done using on resonance RSB

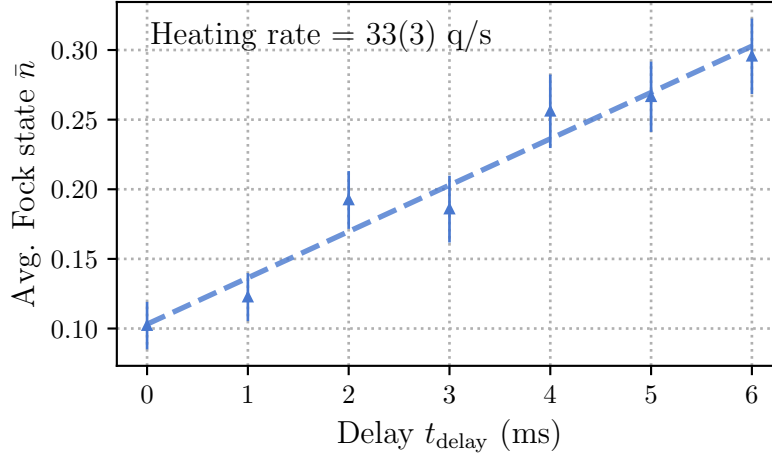


Figure 3.9: Heating rates of upper radial mode, measured by varying the delay between cooling and thermometry pulses. The dashed line is a linear fit to the data, with slope 33(3) quanta/s. The error bars are given by the standard deviation of the fitted $\bar{n}$.

pulses and effective BSB pulses composed of a π -RSB- π pulse sequences. The time dynamics of population flopping is measured as RSB pulse length, t , is varied. In the case of Fock state $|0\rangle$, there should be full contrast “BSB” oscillations, and no visible oscillations when probed with RSB pulses. A thermal Fock state distribution [44] (with truncation at Fock state = 100) is fitted to these signals to extract the mean occupation number, and $\eta\Omega$, the carrier Rabi frequency multiplied by the Lamb-Dicke parameter. A typical thermometry scan after Doppler and sideband cooling can be seen in figure 3.8 *b, c*, where *b* shows the population after effective “BSB” pulses, and *c* shows the population after RSB(t) pulses with duration t . These oscillations are fit to thermal Fock state distributions, shown by the dashed lines. The mean occupation number after sideband cooling is found to be $\bar{n} = 0.03(1)$, and $\eta\Omega = 11.6(1)$ kHz.

3.3.2 Heating Rates

After cooling the motion, the ion will return to a temperature in equilibrium with the environment. However, this process can be slowed by adequate isolation of the ion.

In UHV ion trap systems (see section 2.4), motional heating is predominantly caused by electric field noise from the ion trap [2]. Noise due to thermal charge fluctuations at the trap surface can be mitigated by increasing ion-electrode distances. The *NPL* trap (section 2.2), used here, has an ion-electrode distance of 250 μm , larger than most surface traps, but less than that of a macroscopic blade or rod style trap [45]. Heating rates were probed by a series of thermometry experiments, explained above, with varying delay times, t_{delay} , between cooling and thermometry pulses. A typical plot can be seen in figure 3.9. The heating rate of the system was found by a linear fit, shown by the dashed line, to be 33(3) quanta per second on the upper radial 4 MHz mode on one ion. As will be discussed in section 3.3.4, the heating rate sets a limit for the motional coherence time.

3.3.3 Motional Mode Stability

Due to thermal drifts and microphonics introducing amplitude and frequency noise on the RF chain, drifts of the radial motional mode frequencies are seen over time. As will be discussed in the following section 3.3.4, this can lead to dephasing of the motional state. Here, the current motional mode stability is characterised to estimate its contribution to dephasing, and to set a benchmark for future stability improvements.

The motional mode frequency is measured via an RF “tickle” experiment. This consists of short ($\sim 5 \mu\text{s}$) RF field pulses applied to one of the trap DC electrodes which has a non-zero electric field projection to the mode we wish to probe. The frequency of the RF field is scanned around the expected motional mode frequency, and when resonant, drives the motion of the ion. This resonance is measured by observing an increase in fluorescence counts using a 397-nm laser red-detuned by 100 MHz. To measure mode frequency drift, this sequence is repeated every 10 seconds over multiple hours. A typical plot can be seen in figure 3.10 *a*. Over this two hour period, the frequency had an absolute drift of 1.5 kHz. Typically, we calibrate the mode frequency every 10 minutes, which is indicated with the red

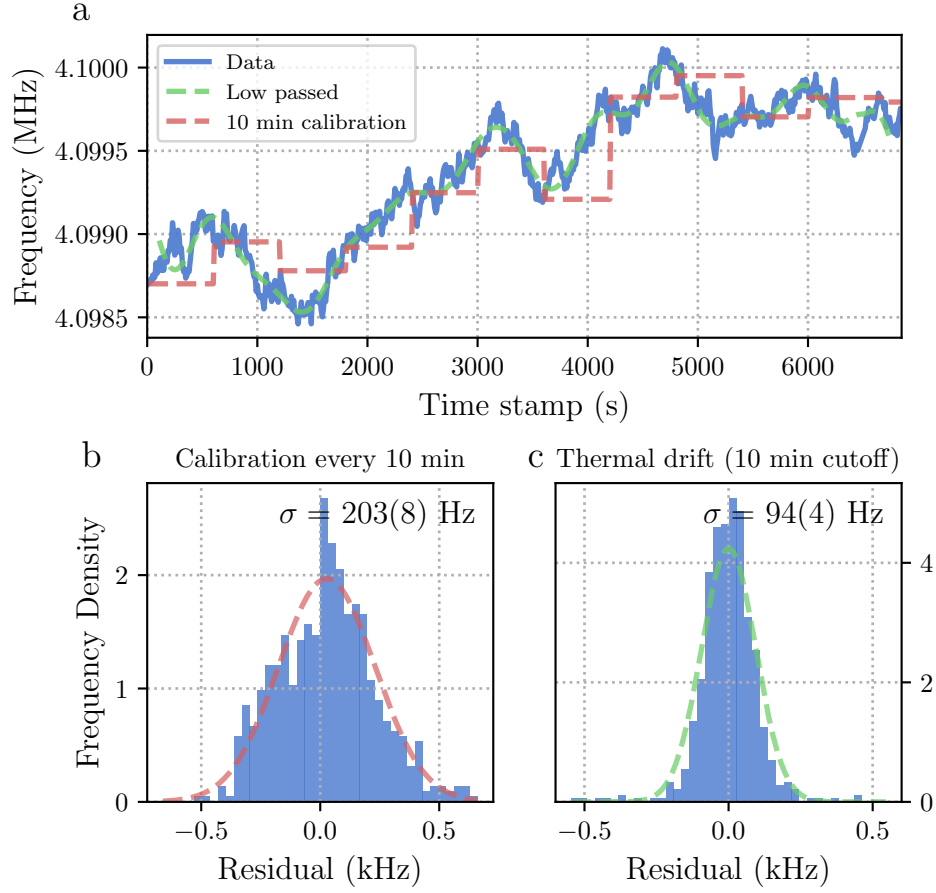


Figure 3.10: **a)** Motional mode frequency drift over time. The radial motional mode frequency is measured by exciting motion using an RF field in resonance with the motional frequency, and looking at increased fluorescence. The resulting frequency trace is shown in blue. The green line shows a low pass filter of the data, with cut-off frequency of 1.67 mHz, corresponding to an assumed 10 minutes slow thermal drift. The motional mode frequency is typically calibrated every 10 minutes, which is indicated by the red line. **b)** Residuals from the measured data and expected calibrated mode frequencies. The standard deviation of these residuals suggests the typical mis-set detuning of the mode frequency in current experiments with automated mode frequency calibrations every 10 minutes. **c)** Residuals from the measured data and the low passed trace. The standard deviation of these residuals are due to noise with period less than 10 minutes.

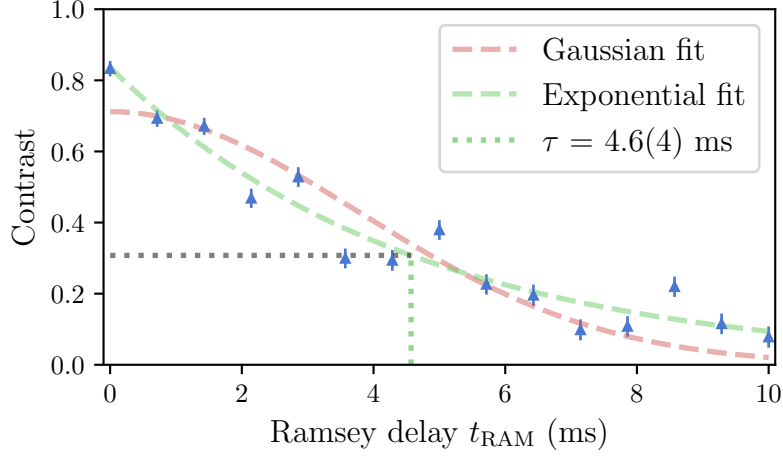


Figure 3.11: Motional coherence time measured by Ramsey sequence on the motional state. The blue points are the population in $|\downarrow\rangle$ after a Ramsey sequence with varying delay time, t_{RAM} . The dashed line is an exponential decay fit, with a coherence time $\tau = 4.6(4)$ ms.

dashed line. Residuals between this calibrated line and the data, shown in *b*, have a standard deviation of $\sigma/(2\pi) = 203(8)$ Hz, giving the expected uncertainty of the mode frequency detuning given our current periodic calibration routine. A slow drift with frequency cut-off of 1.67 mHz is also fitted to the data, shown in green. The residuals from this fit, shown in *c*, have a standard deviation of $\sigma/(2\pi) = 94(4)$ Hz. This suggests that the motional mode frequency is relatively stable over time scales shorter than 10 minutes, but has a slow drift on longer time scales, which is likely due to thermal fluctuations.

3.3.4 Motional Coherence Times

We will utilise the motional modes of the ion to store quantum information. As with the section on spin coherence times 3.2.4, the fidelity of operations, and success probability of algorithms will be limited by the coherence time of the motional state. To measure motional coherence, a Ramsey sequence is performed between states $|\downarrow, 0\rangle$ and $|\downarrow, 1\rangle$, where the first element is the qubit state, and the

second element is the motional Fock state. To prepare these states the following pulse sequence is applied:

1. State prepare system to $|\downarrow, 0\rangle$ via optical pumping and sideband cooling.
2. Apply $R_y(\pi/2)$ -pulse on carrier transition to prepare the qubit in $(|\downarrow, 0\rangle + |\uparrow, 0\rangle)/\sqrt{2}$.
3. Apply $R(\pi)$ -pulse on the RSB to prepare the motional state in $(|\downarrow, 0\rangle + |\downarrow, 1\rangle)/\sqrt{2}$.
4. Delay for some time t_{RAM} .
5. Apply $R(\pi)$ -pulse on the RSB to map accumulated phase onto the qubit state.
6. Apply $R_\phi(\pi/2)$ -pulse on the carrier, with variable axis of rotation, ϕ .
7. Measure the population in $|\downarrow\rangle$ via state-selective fluorescence, section 3.2.2.
8. Repeat steps 1-7 for different values of ϕ to find the contrast of the Ramsey fringe.
9. Repeat steps 1-8 for different values of t_{RAM} to find the decay in contrast with time. The coherence time is defined as the time at which the contrast has decayed to $1/e$ of its initial value.

There are two main mechanisms for motional decoherence[46]: motional heating, as characterised in section 3.3.2, and motional dephasing due to mode frequency instability, as discussed in section 3.3.3.

A heating rate dominated coherence time is given by, $\tau_{\text{HEAT}} = (\sqrt{e} - 1)/\dot{n}$ [46], where $\dot{n}$ is the heating rate. A dephasing dominated coherence time, if noise is correlated on timescales long compared to the Ramsey delay [47], is characterised by a Gaussian decay profile with a coherence time of $\tau_{\text{DEPH}} = \sqrt{2}/\sigma_\omega$.

Evaluating both of these models with $\dot{n} = 33(3)$ q/s, and $\sigma_\omega/(2\pi) = 94(4)$ Hz, the expected motional coherence times of $\tau_{\text{HEAT}} \approx 20$ ms and $\tau_{\text{DEPH}} \approx 2.3$ ms respectively, are found. From these auxiliary characterisations of the motional

modes, it is expected that the coherence time will be limited by motional dephasing. Figure 3.11 shows the decaying coherence when probed by the above sequence. It can be seen that the contrast has a good fit assuming an exponential decay (green dashed line), defined by $C(t_{\text{RAM}}) = C(0)e^{-t_{\text{RAM}}/\tau}$, where $C(t_{\text{RAM}})$ is the contrast at time t_{RAM} , and τ is the coherence time. The fit yields a motional coherence time of $\tau = 4.6(4)$ ms, defined as the time at which the contrast decays to $1/e$ of its initial value. An exponential decay suggests that the noise source causing the dephasing is correlated only for durations much shorter than the Ramsey delay times [47]. The Gaussian decay, (red dashed line), which was appropriate for the previous spin coherence time measurements in section 3.2.4, is a poor fit here.

3.4 Spin-Motion

For full control of the spin-motion hybrid system, we require interactions that couple the two. Perhaps the simplest of this class of interactions are the red- and blue-sidebands (RSB and BSB respectively). The RSB interaction was used previously in the thermometry and sideband cooling sections 3.3.1. The RSB (BSB) consists of a single frequency laser tuned at the carrier frequency minus (plus) the motional mode frequency, ω_m . The interaction is well described by the (Anti-) Jaynes-Cummings Hamiltonian, and effectively couples spin flips with the addition or subtraction of a motional quanta depending on the initial spin state. Here, another such interaction coupling spin and motion, known as the spin-dependent force, SDF, is introduced. While the RSB and BSB interactions couple the spin and motion in the motional Fock basis, the SDF couples them through spin-dependent displacements, thus in a coherent state basis. The direction of the displacement is determined by the spin state and an effective detuning parameter.

3.4.1 Spin-Dependent Forces

The optical Mølmer-Sørensen (MS) scheme [6] is used to generate the SDF via a bichromatic laser field. Bichromatic refers to the simultaneous application of two

tones symmetrically detuned around the qubit carrier frequency, with absolute detuning approximately equal to the motional mode frequency, $\delta \approx \omega_m$. The resulting interaction, when ignoring off-resonant and higher order couplings, is given by,

$$\hat{H}_{MS} = i\hbar\eta\Omega\hat{\sigma}_\phi \cos(\delta t) \left(\hat{a}e^{-i\omega_m t} + \hat{a}^\dagger e^{i\omega_m t} \right), \quad (3.5)$$

where η is the Lamb-Dicke parameter, Ω is the carrier Rabi frequency, $\hat{a}(\hat{a}^\dagger)$ is the lowering (raising) operator, and $\hat{\sigma}_\phi$ is the Pauli operator with $\hat{\sigma}_\phi = \cos(\phi)\hat{\sigma}_x + \sin(\phi)\hat{\sigma}_y$. Applying the rotating wave approximation, and defining $\delta_g = \delta - \omega_m$, the interaction Hamiltonian can be approximated to,

$$\hat{H}_{MS} \approx \frac{i\hbar\eta\Omega}{2} \hat{\sigma}_\phi \left(\hat{a}e^{-i\delta_g t} + \hat{a}^\dagger e^{i\delta_g t} \right). \quad (3.6)$$

The trajectory of this displacement can be controlled by varying δ_g : on resonance, $\delta_g = 0$, corresponds to linear trajectories, whilst off resonance, $\delta_g \neq 0$, corresponds to cyclic trajectories where after some duration $t = 2\pi/\delta_g$, the motion returns to the initial state (with perhaps some phase shift [48]). The dynamics of the SDF may be measured by projective spin measurements, the population in $|\downarrow\rangle$, $P_\downarrow$, after applying the SDF for duration t is described by [49],

$$P_{\downarrow, \text{th}} = \frac{1}{2} \left[1 + e^{-4(\bar{n} + \frac{1}{2})|\alpha(t, \delta_g)|^2} \right], \quad (3.7)$$

where $\bar{n}$ is the initial thermal phonon occupation, and $|\alpha(t, \delta_g)| = \Omega_{\text{SDF}} \sin(\delta_g t/2)/\delta_g$, is the absolute displacement of the initial state. This control is utilised in both two-qubit entangling gate experiments, as well as in the creation of squeezed states.

Calibrating the SDF

Geometric phase gates, such as the MS interaction, are widely employed in ion trap experiments due to their robustness against variations in initial motional state. However, the SDF is sensitive to various frequency and power miscalibrations and drifts of these between calibrations. Here, we describe the workflow for calibrating and optimising the SDF behaviour.

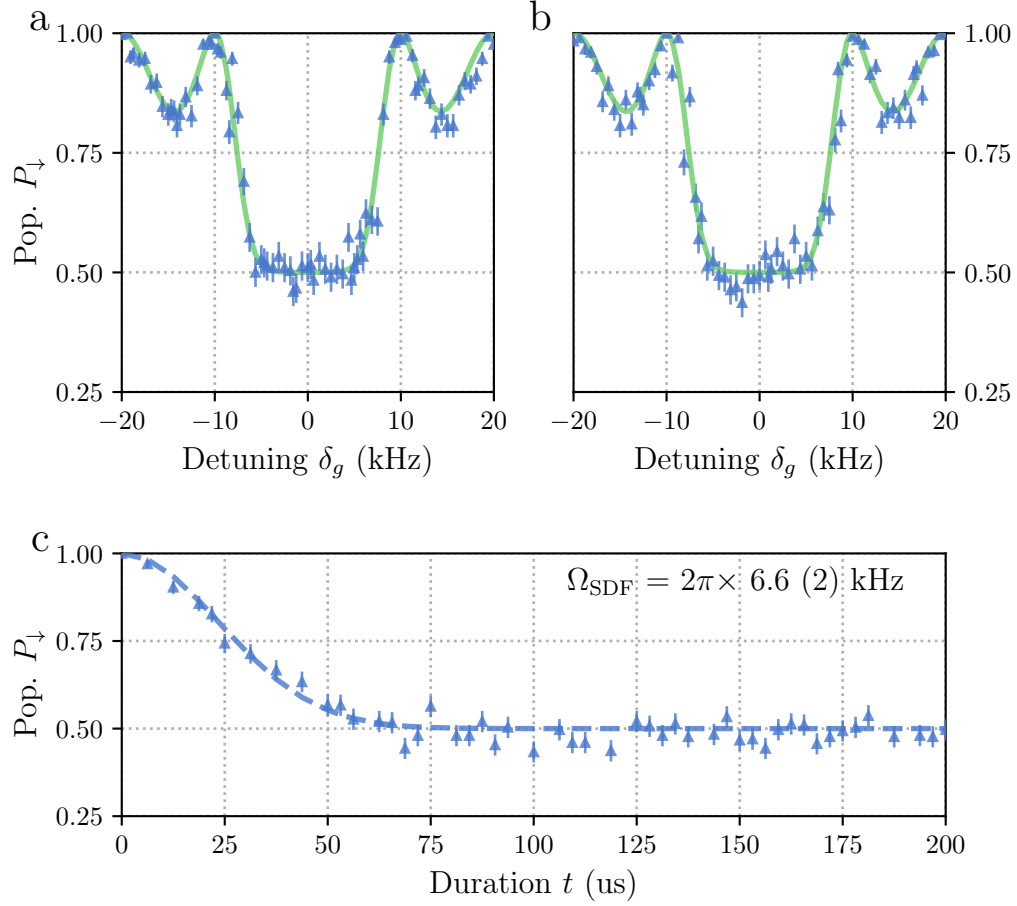


Figure 3.12: Detuning and duration scans of the spin-dependent force applied to an initial thermal state with $\bar{n} = 0.03(1)$. **a)** Detuning scan, where the detuning, δ_g , is varied, and the duration is fixed at $t = 100 \mu\text{s}$. The green line is a fit to the expected behaviour given by equation 3.7. **b)** Detuning scan with a mis-set qubit frequency of 0.5 kHz. The green line is the same fit as in *a* to guide the eye. **c)** Duration scan, where the duration, t , is varied, and the detuning is fixed at $\delta_g = 0$. The dashed line is a fit to the expected behaviour given by equation 3.7 with only Ω_{SDF} as a free parameter.

In practice, the SDF interaction involves relatively few degrees of freedom: the central qubit frequency, the motional mode frequency, the power in each tone, and the durations of the pulses must all be calibrated. The complexity comes from the fact that the ion is a multi-level system, possesses multiple motional modes, and that each experimental parameter can only be controlled with finite precision.

The effect of nearby motional modes is negated by either selecting the interaction mode to be well separated from the others either in frequency, by beam geometry, by operating the SDF near to resonance of the desired mode, or by using amplitude shaping of the pulse. Smoothing a square pulse through amplitude shaping narrows the pulse frequency bandwidth, and thus reduces unwanted off-resonant excitation of other modes.

The power balance of the SDF tones is complicated due to our use of AOMs. AOMs generate frequency shifts in the laser beam at the expense of small frequency-dependent angular shifts. After the AOM, the beam is coupled into a single mode fibre (see figure 2.7), and so the AOMs effectively introduce a frequency-dependent loss. To calibrate the power balance, a pick off of the bichromatic beam after the ion is monitored on a high bandwidth photodiode. The beatnote contrast is measured for the two tones and the power balance is optimised by maximising this contrast. The far off-resonant levels of our ion lead to light-shifts of our qubit frequency. To account for the light-shift the central qubit frequency must be calibrated at the optical power used for the interaction. This is done with the SDF interaction itself. When the qubit frequency is set incorrectly, the two tones will no longer have the same absolute detuning from their respective RSB or BSB. This manifests as a “skewness” of the SDF detuning trace. By varying the qubit frequency and inspecting these detuning scans, the desired SDF behaviour can be found.

To verify the behaviour of the calibrated SDF, the experimental data is compared with theory. Both “detuning” and “duration” scans are used. The “detuning” scan is performed by varying the detuning, δ_g , of the interaction, whilst keeping the SDF duration, t , constant, and vice-versa for the “duration” scan.

Figure 3.12 *c* shows the measured duration scan described by equation 3.7, with

$\delta_g = 0$ and the motional state assumed to be thermal with average Fock state $\bar{n} = 0.03(1)$ which was found previously from thermometry measurements. Here, the displacement, $|\alpha(t)| = \Omega_{\text{SDF}}t/2$, where Ω_{SDF} is the SDF amplitude. The dashed line fit is given by floating Ω_{SDF} only, and finds $\Omega_{\text{SDF}}/(2\pi) = 6.6(2)$ kHz. These measurements were performed with total 729-nm power of 2 mW and $\delta_{\text{LS}}/(2\pi) = 3.5$ kHz, the light shift due to the 729-nm laser relative to the unperturbed qubit frequency.

The detuning scan, shown in figure 3.12 *a*, has the same experimental parameters, but with fixed duration $t = 100$ us. The fit in green is taking the displacement to be $|\alpha(t)| = \Omega_{\text{SDF}} \sin(\delta_g t/2)/\delta_g$, given the value for Ω_{SDF} measured from the duration scan. Here, a qualitatively good agreement between the measured data and the expected behaviour is seen, with the main features being the “closure” at $\delta_g = 2\pi/t$ where the motional state returns to the initial state, and the central flat region around $P_{\downarrow, \text{th}} = 0.5$ where the two motional wave packets are non-overlapping.

Figure *b* shows an example detuning scan when the qubit frequency is mis-set by 0.5 kHz. The central region now displays a clear asymmetry around $\delta_g = 0$. The green line here is the fit from *a* to guide the eye.

3.4.2 Two-Qubit Entangling Gates

We perform two-qubit entangling gates using the Mølmer-Sørensen (MS) interaction [6]. This interaction is the same described SDF from the previous section, but applied globally to two ions. The MS interaction relies on the spin dependent geometric phase accumulated during the motional displacement [40]. To create a two-qubit entangled state, a differential geometric phase of $\pi/2$ must be accumulated between the two-qubit basis states [40]. To ensure there is no residual motional entanglement, the final motional state must return to the initial state. In practise, using an SDF detuned by δ_g , this is achieved by applying the MS interaction for a duration $t_g = 2n\pi/\delta_g$, where n is the number of “loops”. The MS gate is a universal two-qubit gate, and along with only single qubit gates, constitutes a universal gate set for discrete quantum algorithms.

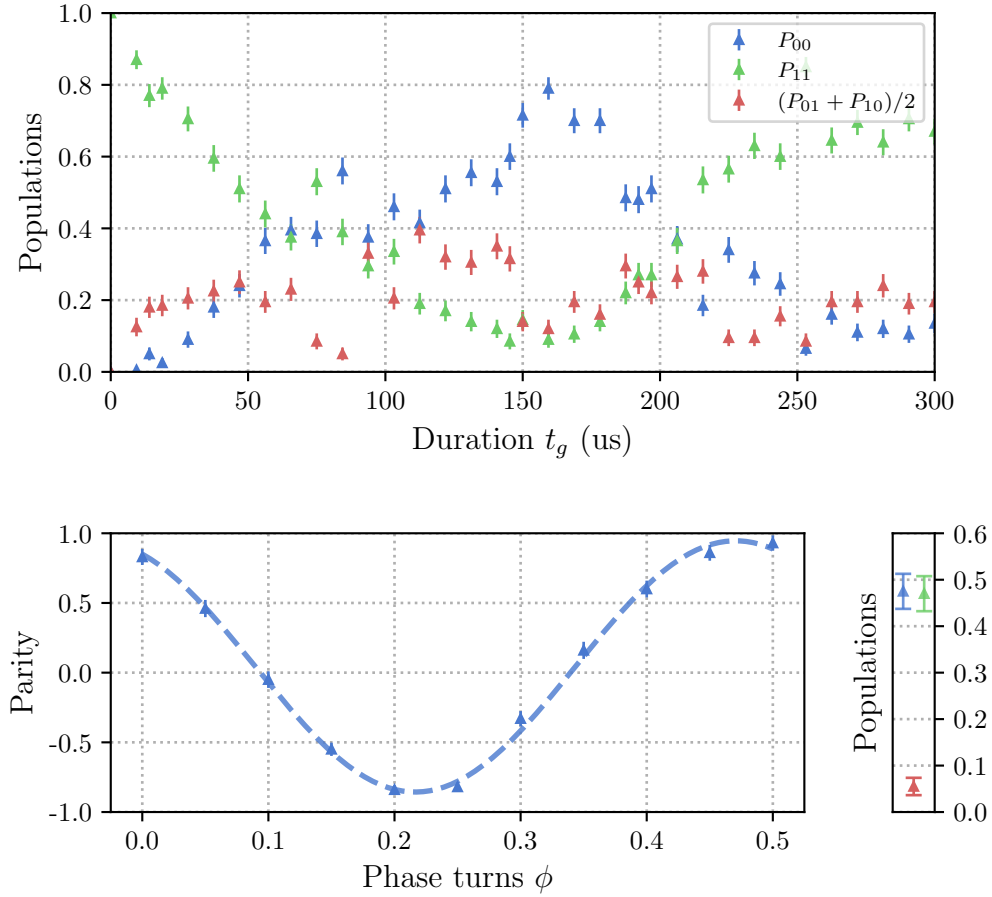


Figure 3.13: Experimental data for $\mathcal{F} = 92(2)\%$ Mølmer-Sørensen (MS) two-qubit entangling gate. **a)** Populations versus duration scan of the MS interaction. Each point corresponds to the average of 200 shots, and the error bars are given by the standard deviation of these averages. The blue points are the populations of both ions being in the $|0\rangle$ state, P_{00} , the green points are the populations of both ions being in the $|1\rangle$ state, P_{11} , and the red points are the ions being the the opposite states, $(P_{01} + P_{10})/2$. The desired entangling gate corresponds here to $t_g = 75\mu\text{s}$. **b)** Parity oscillations after the MS interaction. The parity is defined as $P_{00} + P_{11} - P_{01} - P_{10}$. The fitted contrast here is $0.90(2)$. **c)** Populations at the desired entangling gate duration, $t_g = 75$ us.

Here we quote the current fidelity of experimentally demonstrated two-qubit gates on our system. The fidelity serves to quantify the similarity of two density matrices. Here for a two-qubit entangling gate, we target the creation of the Bell state $|\Phi_{\phi_0}^+\rangle = 1/\sqrt{2}(|00\rangle + e^{i\phi_0}|11\rangle)$, from an initial state of $|00\rangle$, where ϕ_0 is some fixed phase. The fidelity between our mixed state ρ , and the pure Bell state may be given by,

$$\mathcal{F} = \langle \Phi_{\phi_0}^+ | \rho | \Phi_{\phi_0}^+ \rangle = \frac{1}{2}(\rho_{00,00} + \rho_{11,11}) + \frac{1}{2}(e^{i\phi_0}\rho_{11,00} + e^{-i\phi_0}\rho_{00,11}), \quad (3.8)$$

where $\rho_{ij,kl} = \langle ij | \rho | kl \rangle$. To extract the fidelity experimentally, a partial state tomography protocol is used [50], where the first bracketed term of equation 3.8 is measured by performing projective measurements of the two ions after the gate sequence to extract populations, and the second bracketed term, known as the coherence terms, are measured by applying a global analysis $R_\phi(\pi/2)$ pulse to the two ions and applying parity measurements. Parity measurements are defined as $P = P_{00} + P_{11} - P_{01} - P_{10}$, where P_{ij} is the population of the two ions in state $|i\rangle|j\rangle$ after the gate sequence and variable $\pi/2$ -pulse. Creating a Bell state is essential for two-qubit entanglement, but its basis can vary as long as it is consistent across experiments. This allows the Bell state phase, ϕ_0 , to float, which in practice means using only the amplitude of the fitted parity oscillations to compute fidelity, ignoring any phase offset.

The best two-qubit entangling gate fidelity currently achieved on our system is $\mathcal{F} = 92(2)\%$. As shown in figure 3.13, the magnitude of the parity scan was measured to be $0.90(2)$, while the populations $(\rho_{00,00} + \rho_{11,11}) = 0.95(2)$. Each population point in these figures are found by taking the average of 200 shots of the gate sequence.

These results serve as both a proof of principle for full spin control on our system, but also as a benchmark to which we can compare future system improvements. For future work, we will require the use of this entangling gate either as Bell state preparation for input to analogue simulation experiments [20], or as a primitive gate for the spin component of hybrid algorithms [11, 12]. In both these cases, especially

any use that requires multiple concatenated entangling gates, we will likely require improved gate fidelities. Current gate fidelities are limited by the motional dephasing. The gate error due to motional dephasing follows (see section 4.4.6 in [51])

$$\epsilon_{\text{DEPH}} = 0.686 \frac{t_g}{\tau_{\text{DEPH}}}, \quad (3.9)$$

where for these experiments we estimate a motional coherence time of approximately $\tau_{\text{DEPH}} = 1$ ms. Given the gate duration of $t_g = 75$ μs , this gives a dephasing error of $\epsilon_{\text{DEPH}} = 0.05$, or approximately 5%. This is the dominant error source in our current gate fidelity, and can be improved by decreasing the gate duration, or by improving the mode frequency stability. We have not yet attempted to optimise these gates, nor have we attempted to fully account for the observed errors. However, this work serves as a benchmark for future improvements, and a demonstration of the full spin control capabilities of our system.

Appendices



Appendix

A.1 Generating Ions

A.2 Extracting Laser Offset and Magnetic Field

A.3 Experimental Control

A.4 Creating Squeezed States

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